Week 9 — Regression diagnostics & model fit

Business Forecasting

My code (.R) Notes

1 · How far off is the line?

Last week the line gave an answer. This week: how good is that answer, and can we trust it?

The question on the table

Back to Ecobici. In W07 you fitted

$$\widehat{\text{trips}} = 16{,}893 + 723.6 \times \text{temp}$$

on 781 weekdays, and found $R^2 = 0.088$.

The client's finance director reads the report and says:

> Temperature explains 8.8% of ridership. So the model is useless. Throw it away.

Two separate questions hide in that sentence:

1. How well does the line fit — and is the fit real, or luck?
2. Are the assumptions behind the line true? If not, which numbers in the report are wrong?

The first question is model fit. The second is diagnostics. This week is both.

The typical miss

R^2 is a share. The client plans her fleet in trips, so she also needs to know how far off a forecast typically is — in trips.

The residual variance is the average squared miss:

$$s^2 = \frac{\sum_{i=1}^{n} \hat{u}_i^2}{n - 2}$$

Its square root, $s = \sqrt{s^2}$, is the residual standard error. It is measured in the units of y — here, trips.

Exercise 1 — How far off is a typical day?

Use `bikes` — 781 weekdays, with `bikes$trips` (Ecobici trips that day) and `bikes$temp` (average temperature, °C).

1. Regress `bikes$trips` on `bikes$temp`.
2. Compute s^2 and the residual standard error s, in trips.
3. Compute the standard deviation of `bikes$trips`. This is the typical miss if you ignore temperature and always forecast the mean.
4. By how much does the line shrink the typical miss?
5. Find s in `summary(reg)`.

Exercise | Start Over | Show Hint | Run Code

```
1
```

What the typical miss says

	Typical miss (trips)
Always forecast the mean	sd(trips) = 5,548
Forecast from temperature	s = 5,302

- The line barely narrows the forecast: from about 5,550 to about 5,300 trips.
- That matches $R^2 = 0.088$. Both say temperature leaves most of the day-to-day movement unexplained.

Exercise 2 — The landlord's typical miss

The landlord from W07 quotes rents from floor area. The data frame `rents` has 150 of her apartments:

- `rents$size` — floor area, in m^2
- `rents$rent` — monthly rent, in pesos

1. Compute the residual standard error s of the regression of `rents$rent` on `rents$size`, in pesos.
2. Compare it with the standard deviation of `rents$rent`.
3. A tenant asks for a quote. By how many pesos is a quote from size typically off?

Exercise | Start Over | Show Hint | Run Code

```
1
```

Two numbers, two jobs

	R^2	s
What it measures	share of the variation in y the line explains	size of a typical miss
Units	none (0 to 1)	units of y
Ecobici	0.088	5,302 trips
Landlord	0.800	2,708 pesos

R^2 compares models of the same y. s tells a manager how wide to plan around a forecast. You cannot compare 5,302 trips with 2,708 pesos — but you can compare 0.088 with 0.800.

2 · Is the fit real?

A small R^2 could be a weak relationship — or no relationship at all, plus luck. The F-test tells them apart.

A question from a restaurant chain

A restaurant chain orders ingredients from a supplier every day. Some deliveries arrive with ingredients missing.

- x = the hour the order is placed (1 to 24)
- y = the number of missing ingredients in the delivery

Should the chain move its orders to a different hour? Only if the hour really explains part of the mistakes.

That is what an analysis of variance (ANOVA) checks. It compares two things:

- the variation in y that the regression explains — the signal;
- the variation it leaves unexplained — the noise.

ANOVA as a test

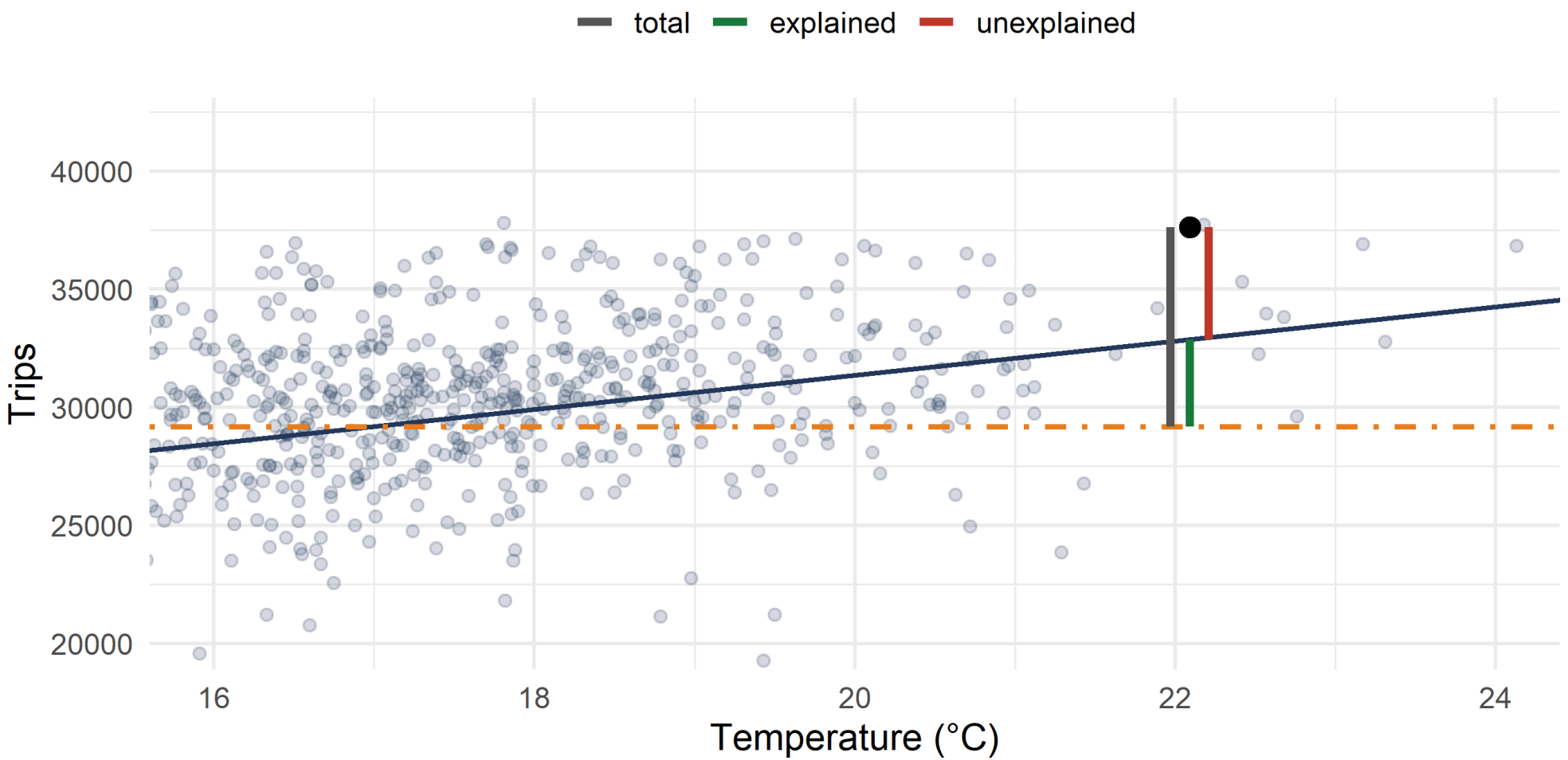
The null hypothesis says the regression explains no more than a model with no predictor at all:

$$H_0 : \beta_1 = 0 \qquad \text{against} \qquad H_A : \beta_1 \neq 0$$

- H_0: the hour does not matter. The best forecast is just the average number of missing ingredients.
- H_A: the hour matters. The regression explains some of the variation.

If we reject H_0, the regression adds value.

One day, three distances



$$\underbrace{y_i - \bar{y}}_{\text{total}} = \underbrace{(\hat{y}_i - \bar{y})}_{\text{explained}} + \underbrace{(y_i - \hat{y}_i)}_{\text{unexplained}}$$

From one day to all days

Square each distance and add over all days:

$$\sum_i (y_i - \bar{y})^2 = \sum_i (\hat{y}_i - \bar{y})^2 + \sum_i (y_i - \hat{y}_i)^2$$

Piece	Formula	Degrees of freedom
SST	$\sum_i (y_i - \bar{y})^2$	$n - 1$
SSR	$\sum_i (\hat{y}_i - \bar{y})^2 = \hat{\beta}_1^2 \sum_i (x_i - \bar{x})^2$	1
SSE	$\sum_i (y_i - \hat{y}_i)^2$	$n - 2$

The degrees of freedom add up too: $(n - 1) = 1 + (n - 2)$.

Good fit, bad fit



- Left: most of the distance from the mean is green. The line explains a lot.
- Right: most of it is red. The line explains little.

The F statistic

Divide each sum of squares by its degrees of freedom. These are the mean squares:

$$MSR = \frac{SSR}{1} \qquad MSE = \frac{SSE}{n-2} = s^2$$

The test statistic compares them:

$$F = \frac{SSR/df_R}{SSE/df_E} = \frac{MSR}{MSE}$$

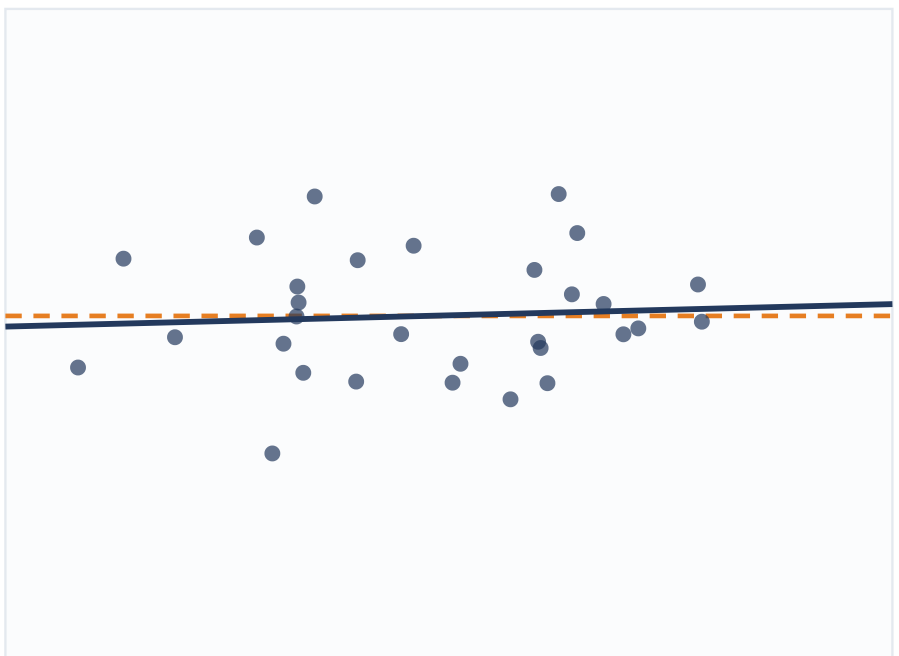
If $H_0 : \beta_1 = 0$ is true, F follows an F distribution with 1 and $n-2$ degrees of freedom. A large F is evidence for $H_A : \beta_1 \neq 0$.

Reject H_0 at level α when $F > F_{1-\alpha;\, 1,\, n-2}$. The p-value is $P(F_{1,\, n-2} \geq F)$. Why the ratio works.

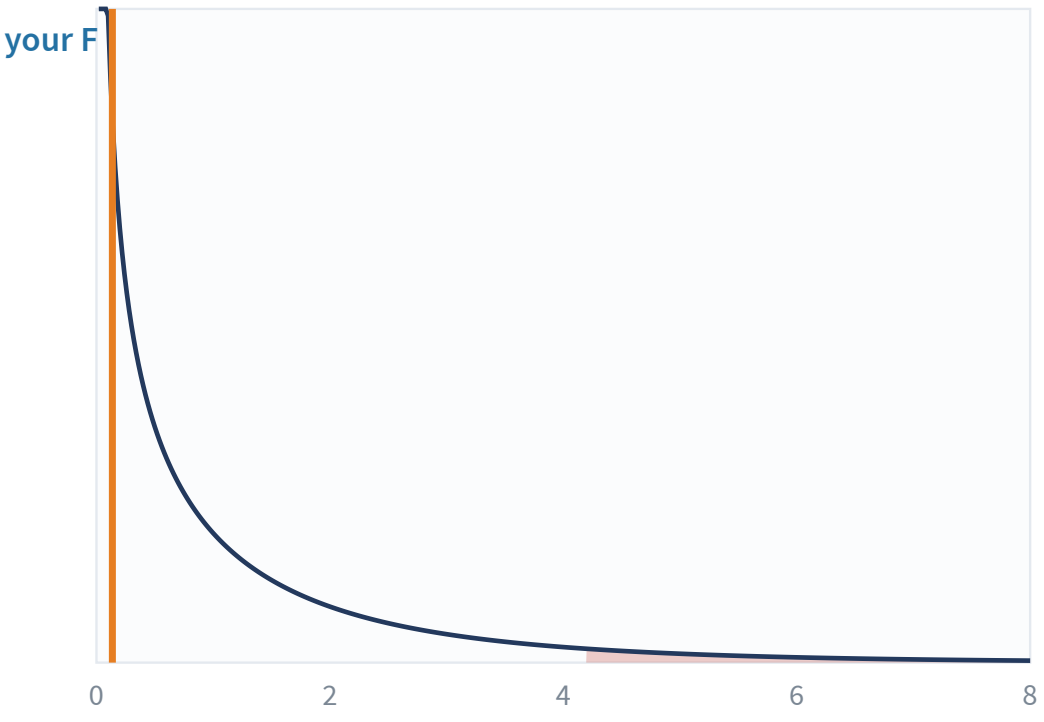
Signal against noise

The test is $H_0 : \beta_1 = 0$ against $H_A : \beta_1 \neq 0$. Left: the data and the fitted line. Right: the F distribution if the true slope were 0, with the 5% rejection region shaded. Move the sliders, then show the numbers.

True slope 0.30 Noise 6.0 n 30 New sample Show the numbers



x (dashed orange = mean of y)



SSR ? SSE ? F = ? 5% critical value ? p-value ?

What to try on it

- Set the true slope to 0. Press New sample a few times. F lands in the unshaded part about 95% of the time.
- Keep the slope small, say 0.3, and raise n. The same weak slope becomes clearly significant.
- Raise the noise. SSE grows, F falls, and the verdict can flip.

A weak relationship measured on many observations can give a large F. That is the Ecobici situation.

Exercise 3 — Is temperature's 8.8% real?

Test at the 5% level whether temperature explains any of the variation in Ecobici trips:

$$H_0 : \beta_1 = 0 \qquad \text{against} \qquad H_A : \beta_1 \neq 0$$

in the regression of `bikes$trips` on `bikes$temp` ($n = 781$).

1. From `anova(reg)`, take SSR and SSE. Build MSR, MSE and the F statistic.
2. Find the 5% critical value and the p-value.
3. Reject or not? Then find your F in the `anova(reg)` output.

Exercise | Start Over | Show Hint | Run Code

```
1 reg <- lm(trips ~ temp, data = bikes)
```

The Ecobici ANOVA table

Source	Sum of squares	df	Mean square	F
Regression (SSR)	2,117,129,482	1	2,117,129,482	75.32
Residual (SSE)	21,895,427,100	n − 2 = 779	28,107,095	
Total (SST)	24,012,556,582	n − 1 = 780		

- `anova(reg)` prints the first two rows. The total row is their sum.
- The mean square of the residuals is s^2; its square root is the 5,302 trips of Exercise 1.

Weak, but real

Question	Statistic	Ecobici
How much does temperature explain?	R^2	0.088
Is that more than luck?	F (5% critical value 3.85)	75.32

Temperature explains little of ridership. But what it explains is not luck: with 781 days, a slope of 724 trips per degree cannot come from a world where $\beta_1 = 0$.

So the finance director is half right. The model is a poor forecaster of any single day. It is not useless: the temperature effect is real.

Exercise 4 — Influencer marketing

A brand pays influencers for product placement. It checks whether placements with more popular influencers bring more visits to its website. A simple regression on 15 campaigns gives, from the ANOVA table, $SSE = 52$ and $SST = 152$.

Test the significance of the regression at the 5% level with the F statistic. Write out the whole test: hypotheses, rejection region, test statistic, conclusion. Use 4 decimal places.

Exercise | Start Over | Show Solution | Run Code

```
1
```

The same test, seen as two models

The F statistic can be written with SST and SSE only:

$$F = \frac{SSR/df_R}{SSE/df_E} = \frac{(SST - SSE)/(df_T - df_E)}{SSE/df_E}$$

- Full model, with x: $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$. What it leaves unexplained is SSE.
- Restricted model, intercept only: $\hat{y}_i = \bar{y}$. What it leaves unexplained is SST.

The top of F measures how much the unexplained part shrinks when we move from the restricted model to the full one. If it shrinks a lot, reject the restricted model.

With one regressor this is exactly the ANOVA test. From W10 on, with several regressors, the same idea tests whether a group of extra predictors helps.

Three tests, one answer

With a single regressor, three tests ask the same question:

Test	Null against alternative	Statistic
correlation test	$H_0 : \rho = 0$ against $H_A : \rho \neq 0$	$T = \dfrac{r\sqrt{n-2}}{\sqrt{1-r^2}} \sim t_{n-2}$
t-test on the slope (W08)	$H_0 : \beta_1 = 0$ against $H_A : \beta_1 \neq 0$	$T = \dfrac{\hat{\beta}_1}{SE(\hat{\beta}_1)} \sim t_{n-2}$
F-test	$H_0 : \beta_1 = 0$ against $H_A : \beta_1 \neq 0$	$F = \dfrac{MSR}{MSE} \sim F_{1,\, n-2}$

The correlation test is on your formula sheet. Check the claim on the Ecobici data.

Exercise 5 — Do the three agree?

For `bikes$trips` and `bikes$temp` ($n = 781$):

1. Test $H_0 : \rho = 0$ against $H_A : \rho \neq 0$ at 5%: build T from the correlation, then the two-sided p-value.
2. Square T. Compare it with the $F = 75.32$ of Exercise 3.
3. In `summary(reg)`, find the `t value` of `temp` and the `F-statistic` line.

Exercise | Start Over | Show Hint | Run Code

```
1
```

Back to the restaurant

- The chain regresses missing ingredients on the hour of the order and reads the ANOVA table.
- A large F (small p-value) means the hour really does explain part of the mistakes. The chain can move its orders to a better hour.
- A small F means any pattern in the hours is noise. Moving the orders would not help.

Model fit in one table

Quantity	Meaning	Ecobici
SST	total variation in y	2.401×10^{10}
SSR	variation explained by the line	2.117×10^{9}
SSE	unexplained (residual) variation	2.190×10^{10}
$R^2 = SSR/SST$	share of the variation explained	0.088
$s = \sqrt{SSE/(n-2)}$	typical miss, in units of y	5,302 trips
$F = MSR/MSE$	is SSR large compared with SSE?	75.3

The fit is settled for Ecobici: weak, but real. The next question is whether the assumptions behind the line hold.

3 · What the residuals should look like

The errors are never seen. The residuals are our only view of them.

The assumptions, again

1. Linear in the parameters, with an additive error: $y_i = \beta_0 + \beta_1 x_i + u_i$.
2. The error has mean zero: $E(u_i) = 0$.
3. Constant variance: $\mathrm{Var}(u_i) = \sigma^2$.
4. Errors uncorrelated across observations: $\mathrm{Cov}(u_i, u_j) = 0$.

One more is needed for t-tests, F-tests and intervals in small samples:

5. The error is normal: $u_i \sim N(0, \sigma^2)$.

Which tool checks which

Assumption	Look at	A healthy picture
linearity	residuals against fitted values	no curve, no arch
constant variance	residuals against fitted values	a band of even width
uncorrelated errors	residuals in time order; $\hat{u}_i$ against $\hat{u}_{i-1}$	no runs, no tilt
normality	histogram, Q-Q plot, Jarque–Bera test	bell shape, straight line, small JB

The first three are checked with pictures. Normality has a picture and a formal test.

Exercise 6 — A model you can trust?

Use `rents` (`rents$size` in m^2, `rents$rent` in pesos a month).

1. Regress `rents$rent` on `rents$size`.
2. Plot the residuals against the fitted values.
3. Describe the picture. Do the residuals curve? Does their spread change from small to large fitted rents?

Exercise | Start Over | Show Hint | Run Code

```
1
```

A healthy residual plot



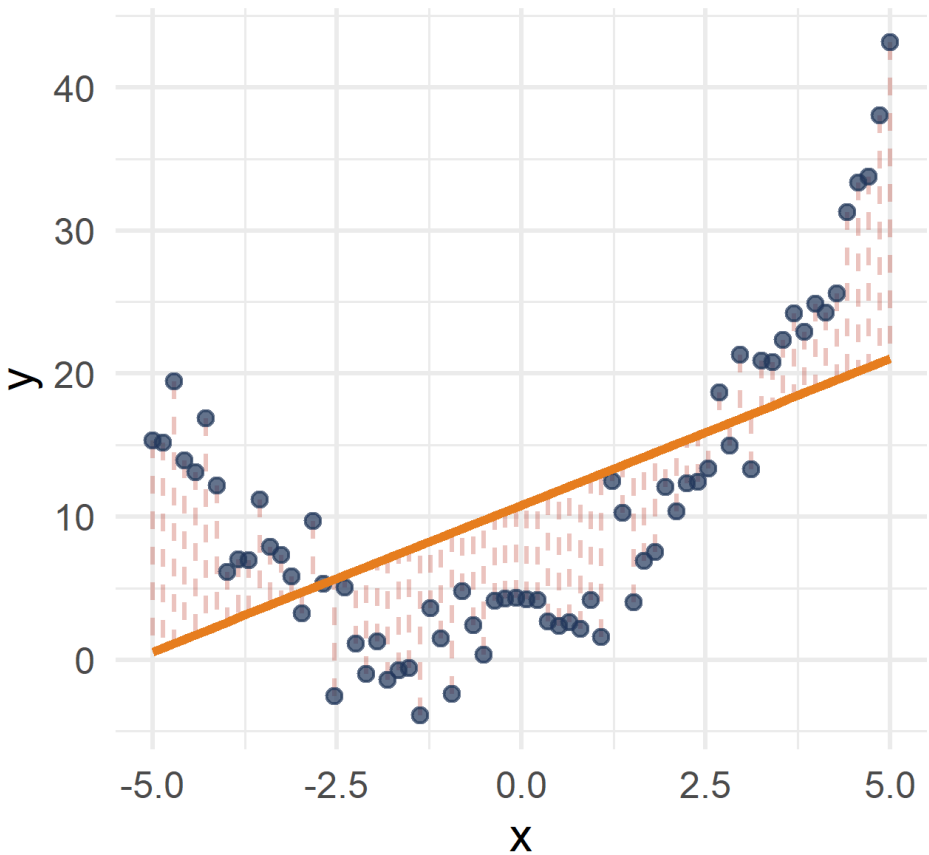
- No pattern: the line is not too high at one end and too low at the other.
- An even band: a quote for a large flat is as reliable, in pesos, as one for a small flat.
- This is the benchmark. Every other plot this week is compared with it.

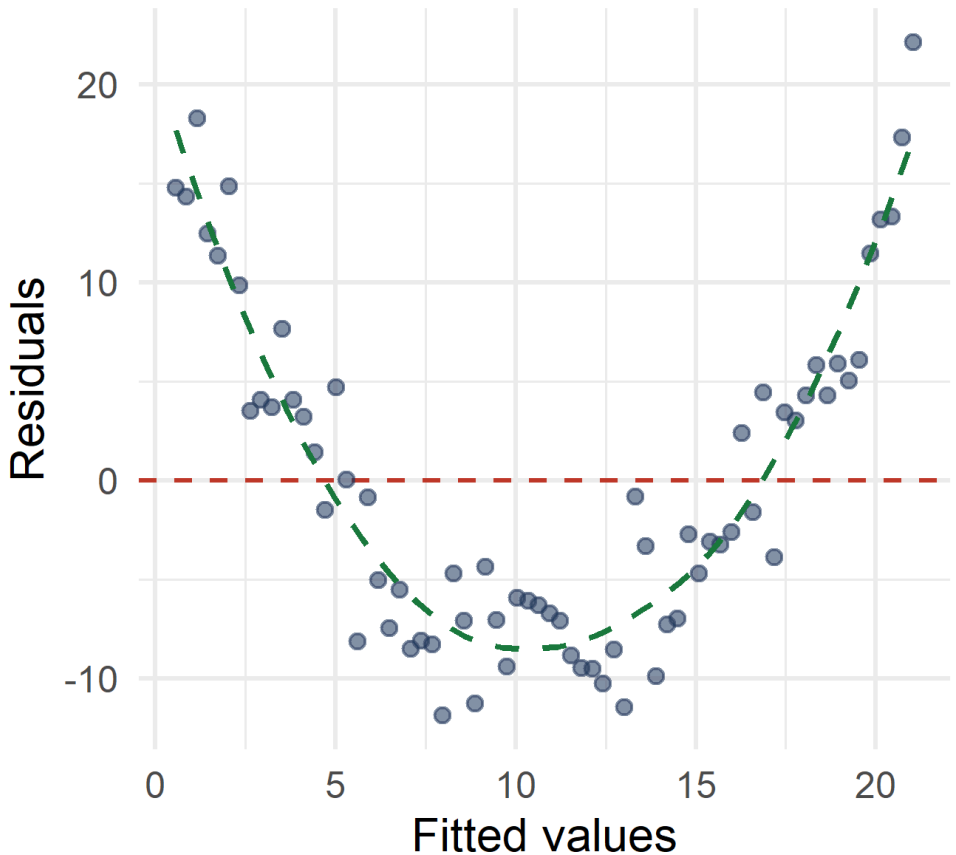
4 · Is a straight line the right shape?

Assumption 1: linearity. A curve fitted with a straight line leaves a curve in the residuals.

Linearity check: curved data

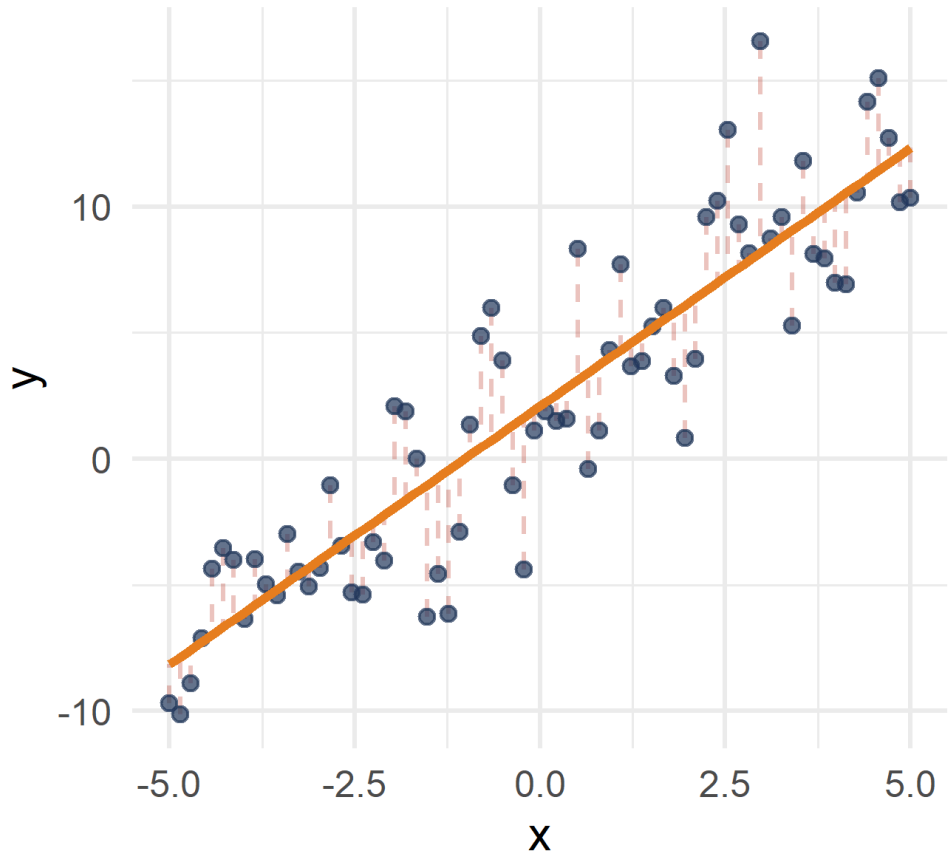
To check linearity, plot the residuals against the fitted values $\hat{y}$ and look for a systematic curve.

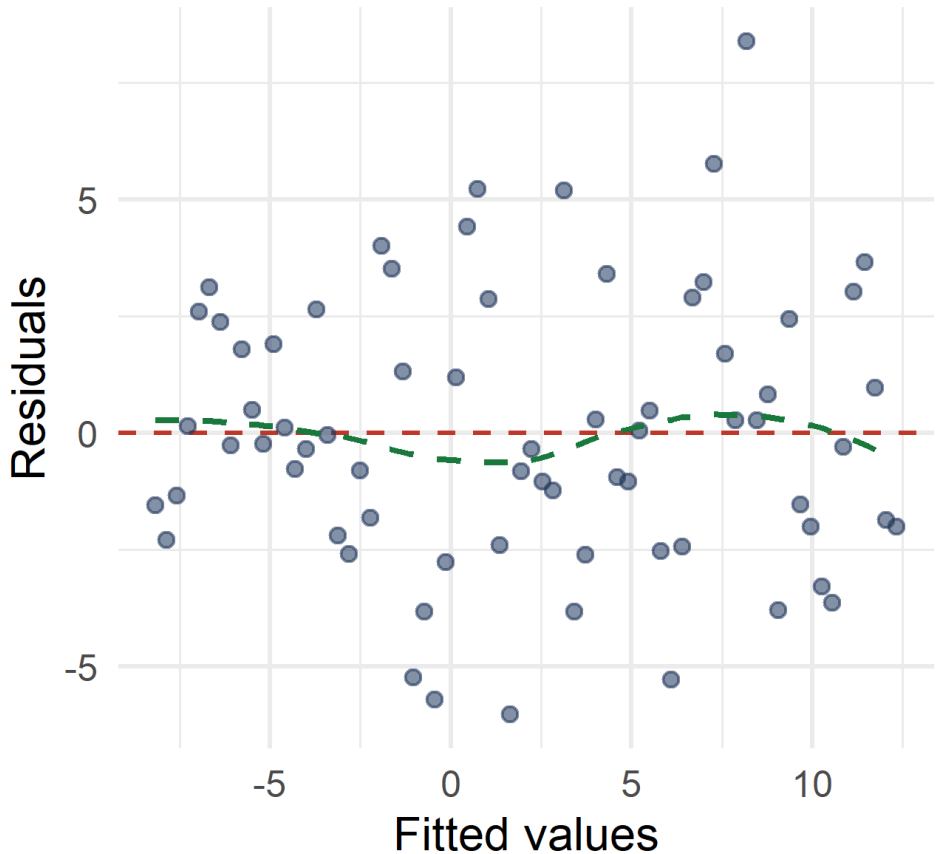




- Left: the line is too high in the middle and too low at both ends.
- Right: the residuals make a U. The dashed green curve follows it.

Linearity check: straight data





- Same kind of data, but the truth is a straight line.
- The residuals scatter around zero with no shape. The green curve stays flat.

The ad budget question

An online furniture retailer ran one month of digital ads in 120 Mexican cities. Each city got a different budget.

The data frame `campaigns`:

- `campaigns$ad_spend` — ad budget that month, in thousands of pesos
- `campaigns$sales` — sales that month, in thousands of pesos

Marketing wants one number: how many pesos of sales does one more peso of ads bring? If that number is the same at every budget, the answer is simple — raise every budget.

Exercise 7 — Fit, then look at the residuals

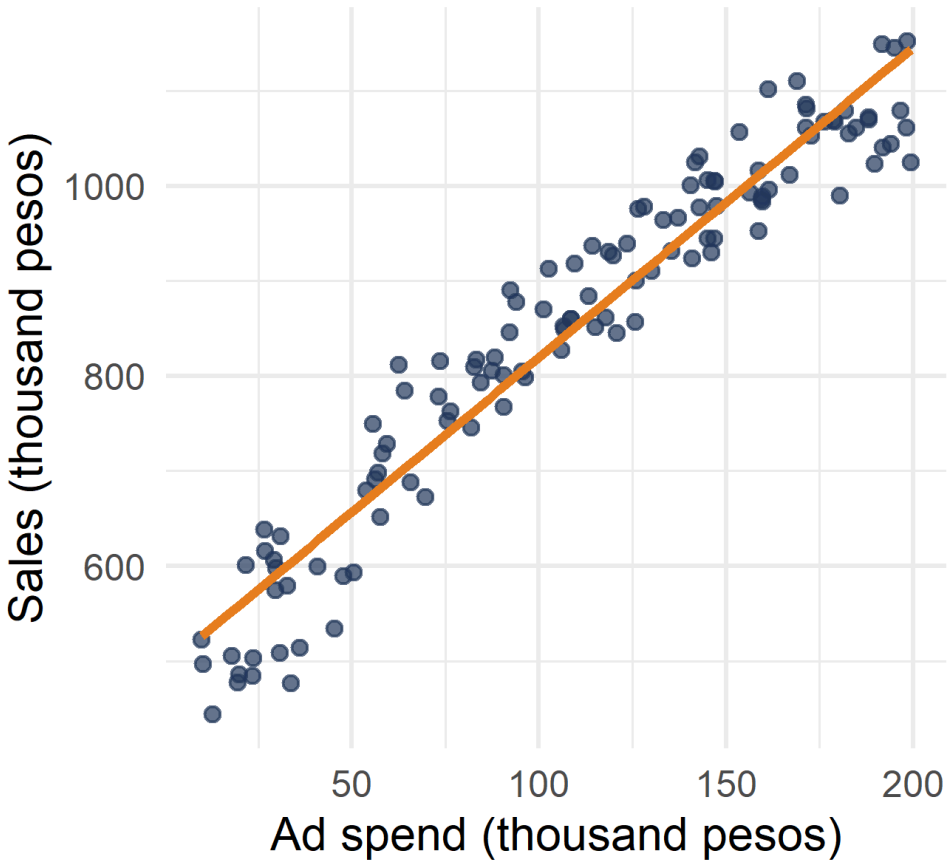
Use `campaigns` (`campaigns$ad_spend` and `campaigns$sales`, both in thousands of pesos).

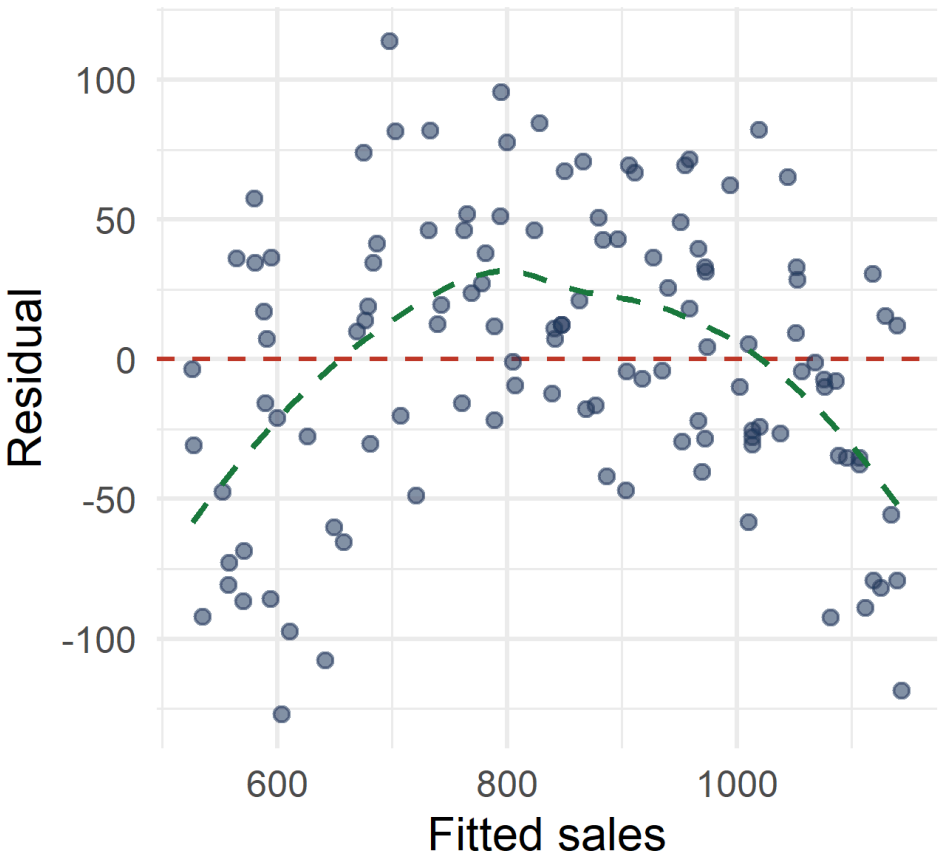
1. Regress `campaigns$sales` on `campaigns$ad_spend`. Read the slope and R^2.
2. Plot the residuals against the fitted values.
3. Compute the mean residual for three groups of cities: budget under 50, between 50 and 150, and over 150. Is the line too high or too low in each group?

Exercise | Start Over | Show Hint | Run Code

```
1
```

An arch in the residuals





- The points bend. The straight line cuts across the bend.
- $R^2 = 0.93$, and the line is still the wrong shape. A high R^2 does not show that the model is right.

🖊️ Exercise 8 — Is one more peso worth the same everywhere?

Use `campaigns` again.

1. Regress `campaigns$sales` on `campaigns$ad_spend` using only the cities with a budget under 60.
2. Do the same using only the cities with a budget over 140.
3. Compare the two slopes with the slope for all cities, 3.26. What does one more thousand pesos of ads bring in a small-budget city? In a large-budget city?

Exercise | Start Over | Show Hint | Run Code

```
1
```

Bend the line

The returns fall as the budget grows. One way to let the line bend is to regress sales on the square root of ad spend:

$$\text{sales}_i = \beta_0 + \beta_1 \sqrt{\text{ad_spend}_i} + u_i$$

The model is still linear in β_0 and β_1 — assumption 1 only asks for that. What changed is the regressor.

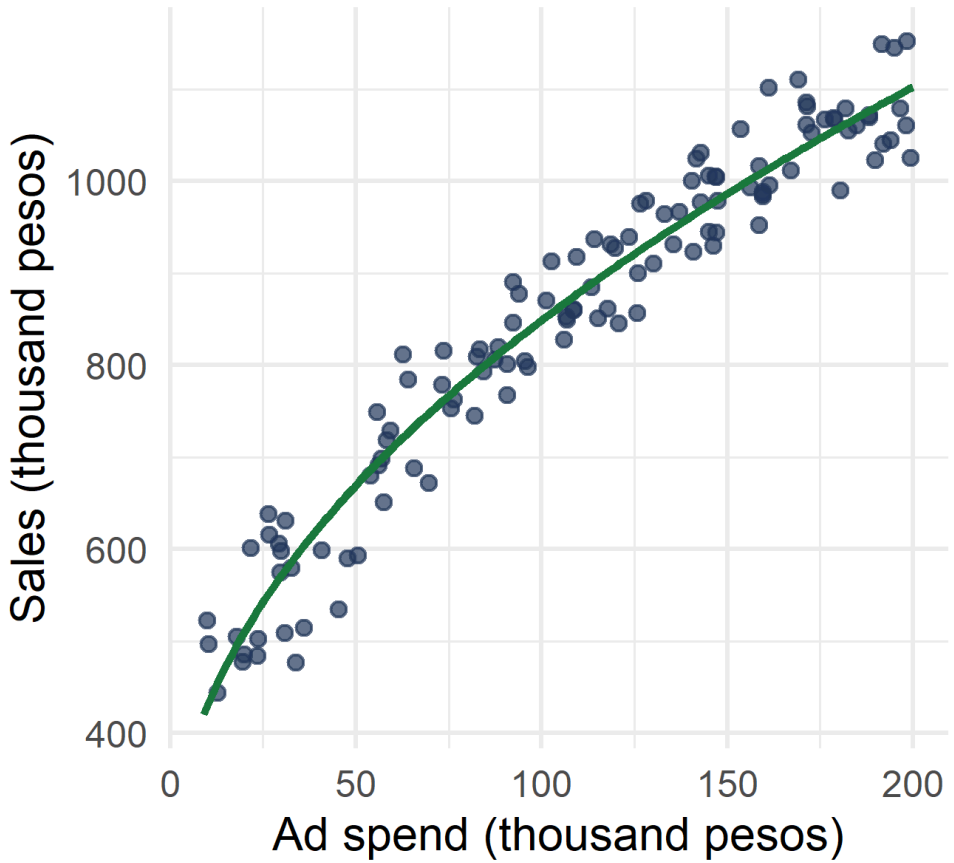
Exercise 9 — Does the bend fix it?

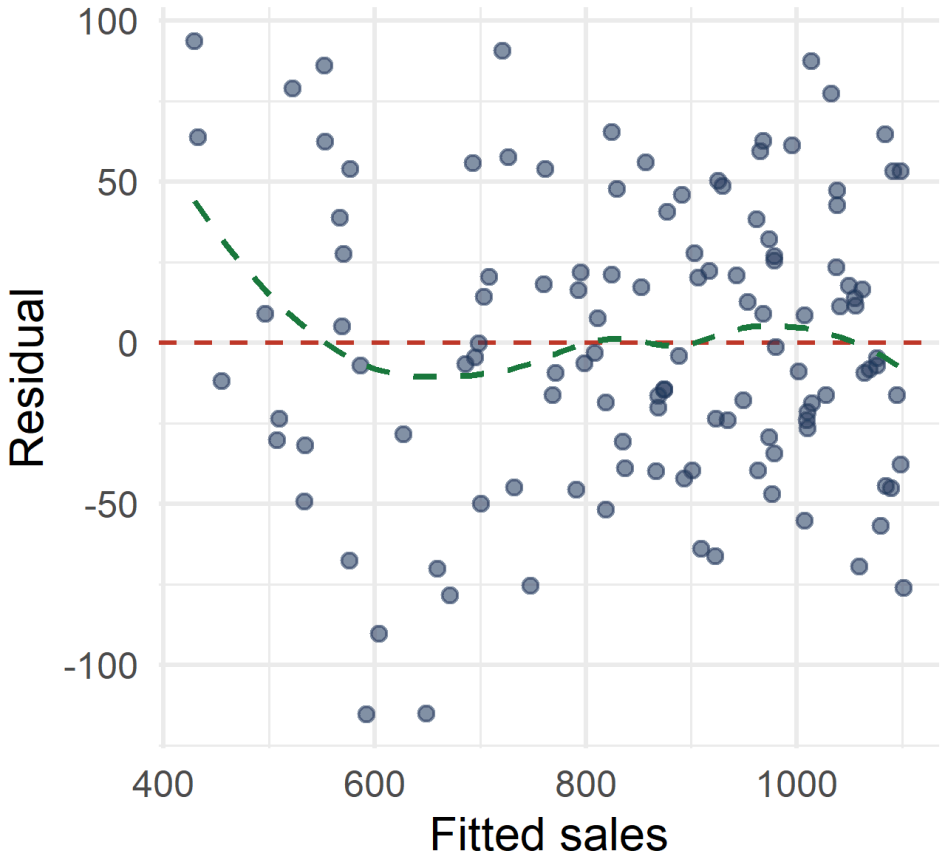
1. Create `campaigns$root_ad`, the square root of `campaigns$ad_spend`.
2. Regress `campaigns$sales` on `campaigns$root_ad`. Read R^2.
3. Plot the residuals against the fitted values, and compute the mean residual in the same three budget groups as Exercise 7. Is the arch gone?

Exercise | Start Over | Show Hint | Run Code

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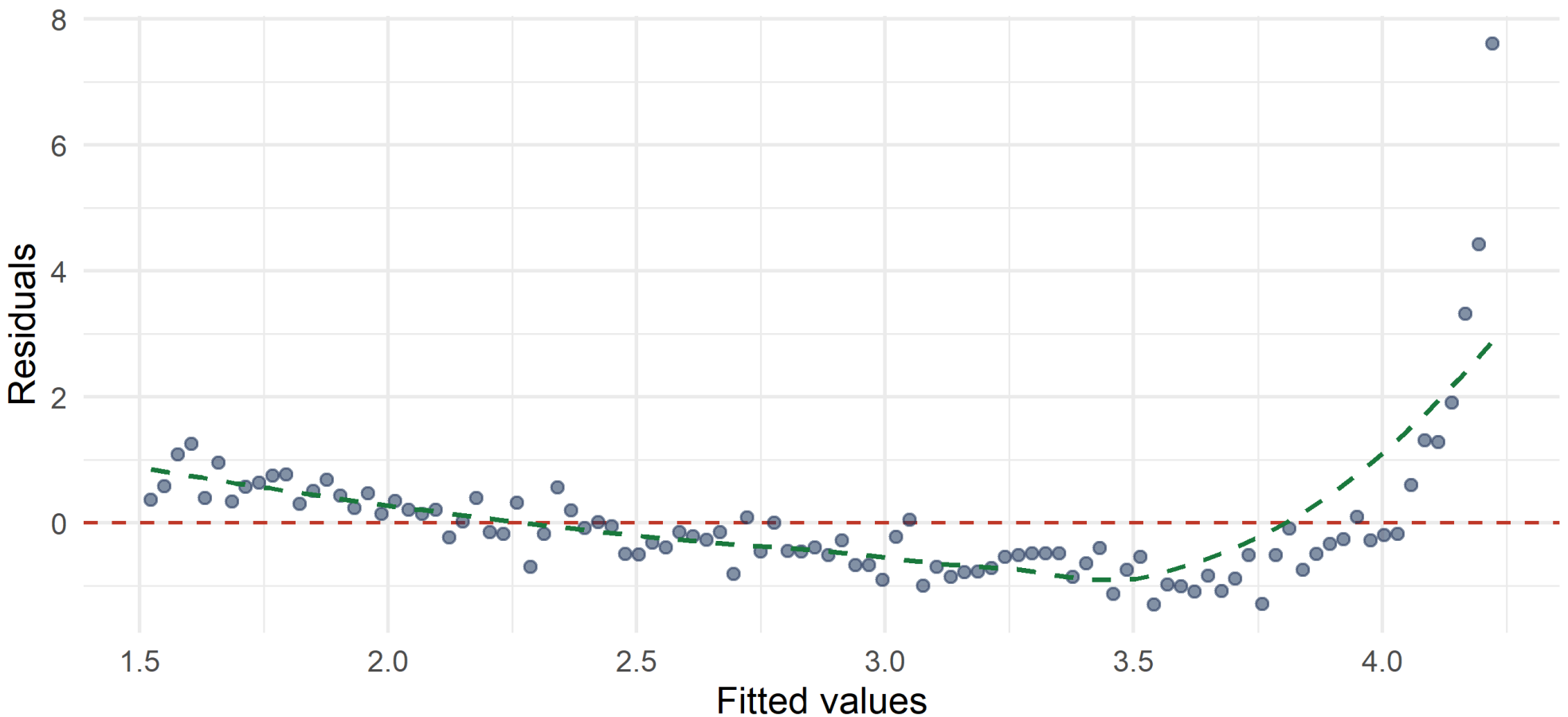
The curve, fitted





- $\widehat{\text{sales}} = 236.3 + 61.24\sqrt{\text{ad_spend}}.$
- At a budget of 20 thousand, one more peso of ads brings about 6.8 pesos of sales. At 180 thousand, about 2.3.

What goes wrong here? — a residual plot



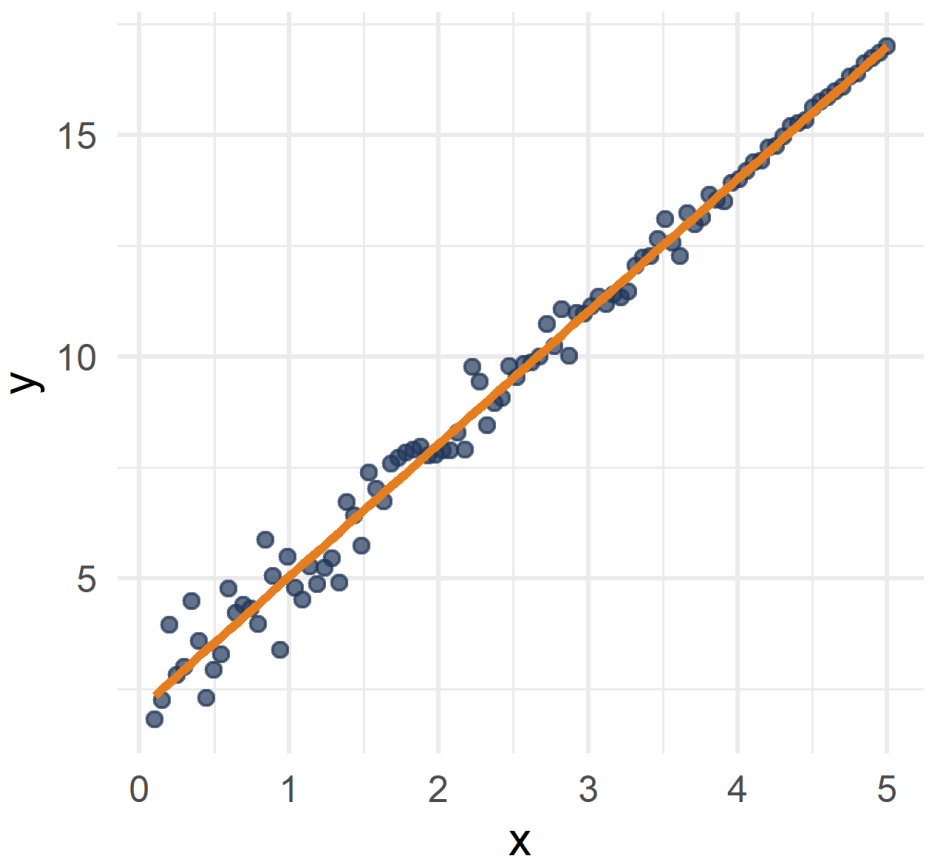
Linearity fails. The residuals drift down, then shoot up at the largest fitted values: the true relationship is a curve that drops fast for small x and then levels off (here y depends on $1/x$). A straight line cannot follow it.

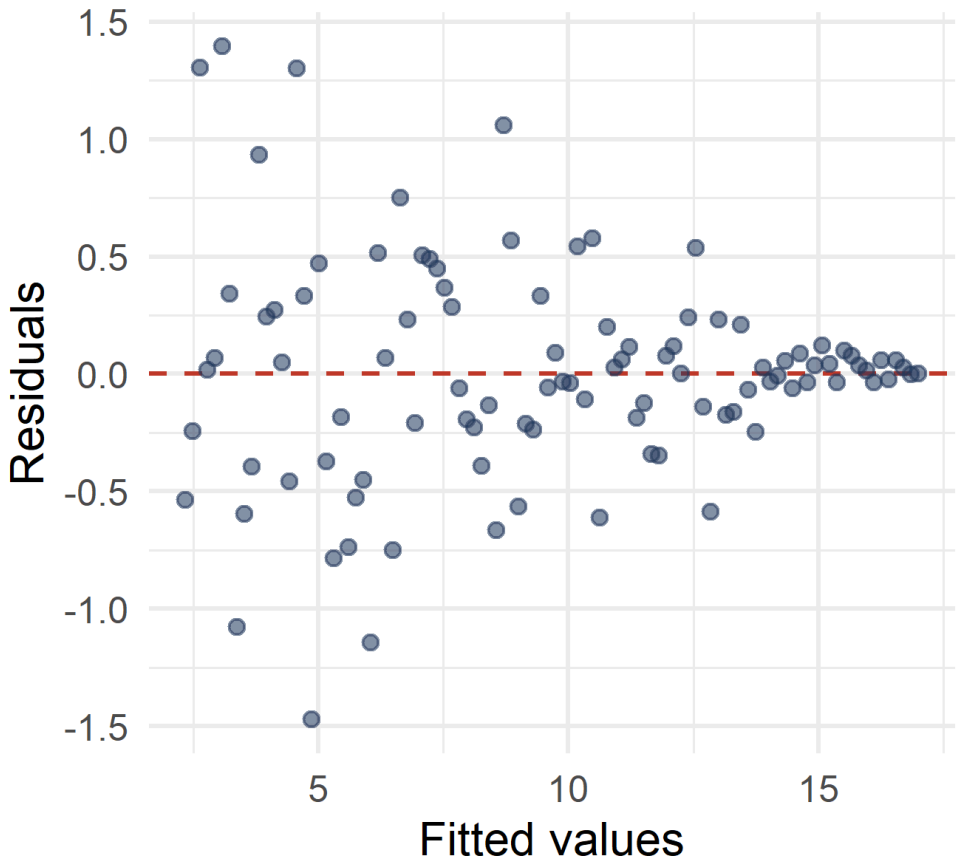
5 · Is the spread constant?

Assumption 3: the same variance everywhere. When it fails, the residuals fan out.

Constant variance: a heteroskedastic example

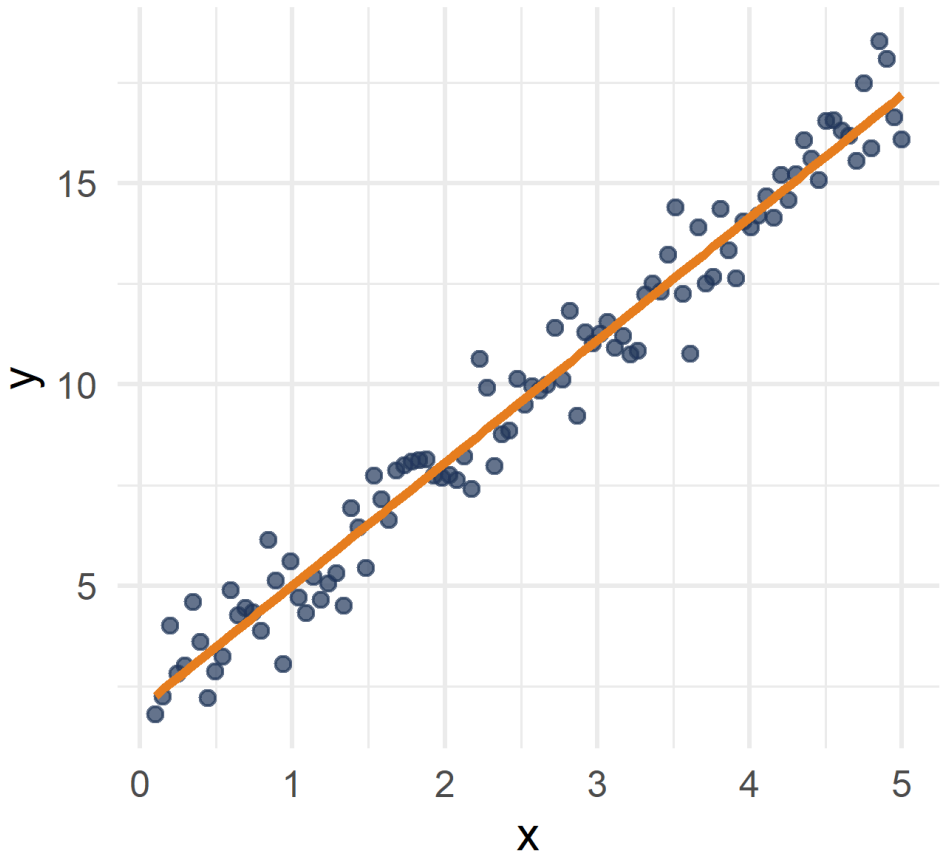
Plot the residuals against $\hat{y}$ and ask: does the vertical spread change as $\hat{y}$ changes?

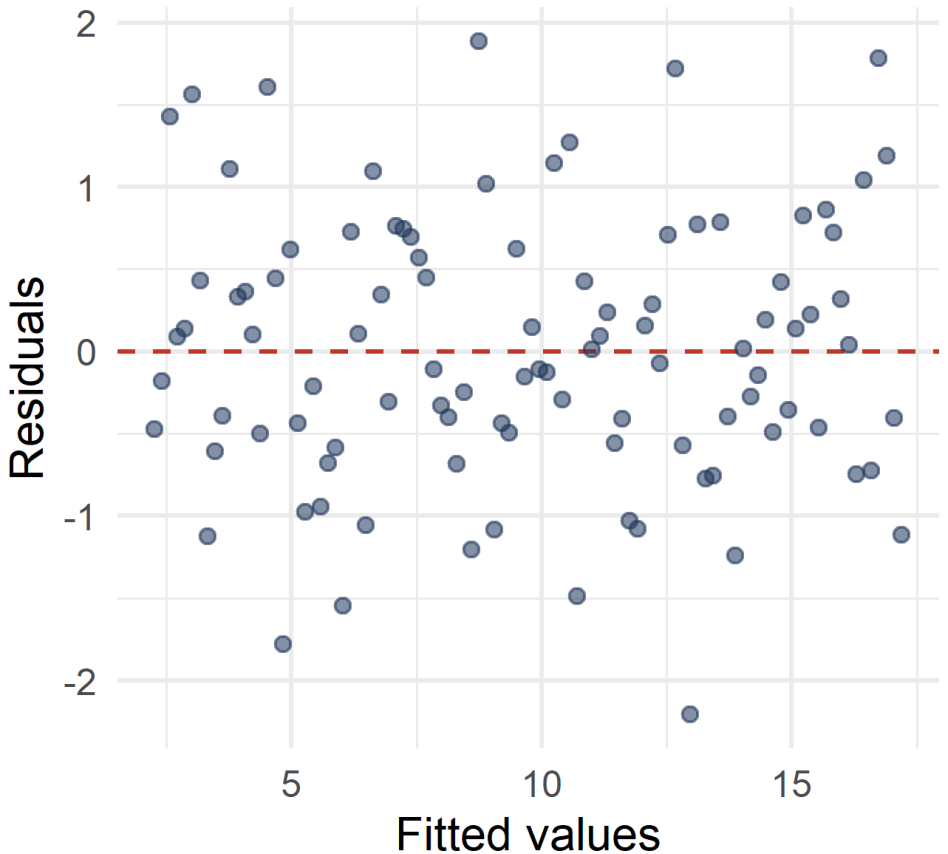




The spread shrinks as $\hat{y}$ grows. The variance of the error is not constant. This is called heteroskedasticity.

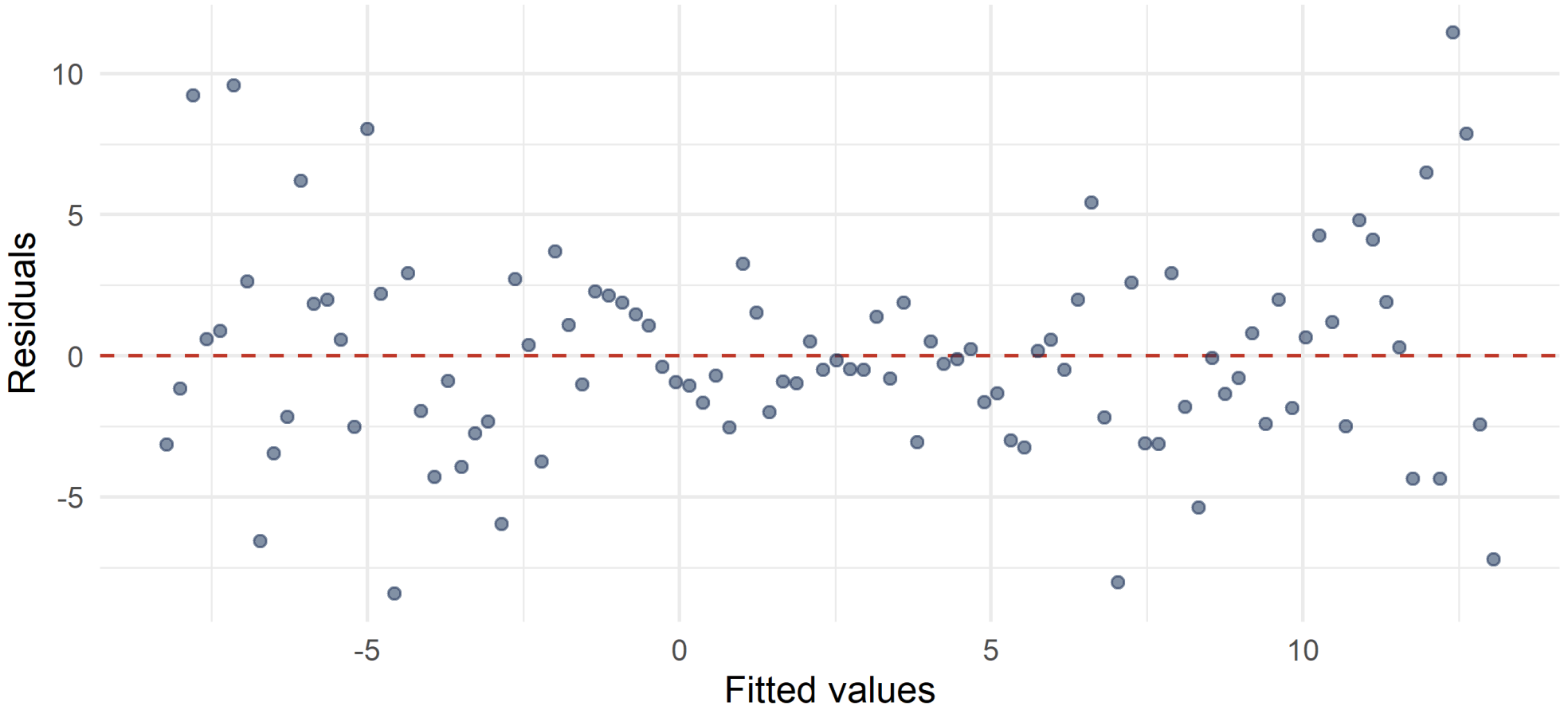
Constant variance: a homoskedastic example





Same line, same x. Now the band has the same width everywhere: homoskedasticity.

What goes wrong here? — another residual plot



Constant variance fails. The spread is wide at both ends and narrow in the middle — a bow tie. The line itself is fine; the scatter around it is not the same everywhere.

Food spending and income

`spending` holds 10,000 Mexican households from the ENIGH 2018 survey, per person, per month, in pesos:

- `spending$food_pc` — food spending
- `spending$income_pc` — income

A supermarket chain forecasts food spending from income to plan its stock and its promotions. It sets a band of plus or minus one typical miss around each forecast. Should that band be the same width for a poor household and for a rich one?

Exercise 10 — Look at the spread

Use `spending` (`spending$food_pc` and `spending$income_pc`, pesos per person per month).

1. Regress `spending$food_pc` on `spending$income_pc`. Read the slope.
2. Plot the residuals against the fitted values.
3. Describe the cloud. Where is it narrow, and where is it wide?

Exercise | Start Over | Show Hint | Run Code

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1
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Exercise 11 — Put a number on the fan

A picture can mislead. Measure it: compute the standard deviation of the residuals within each income quartile of `spending$income_pc`.

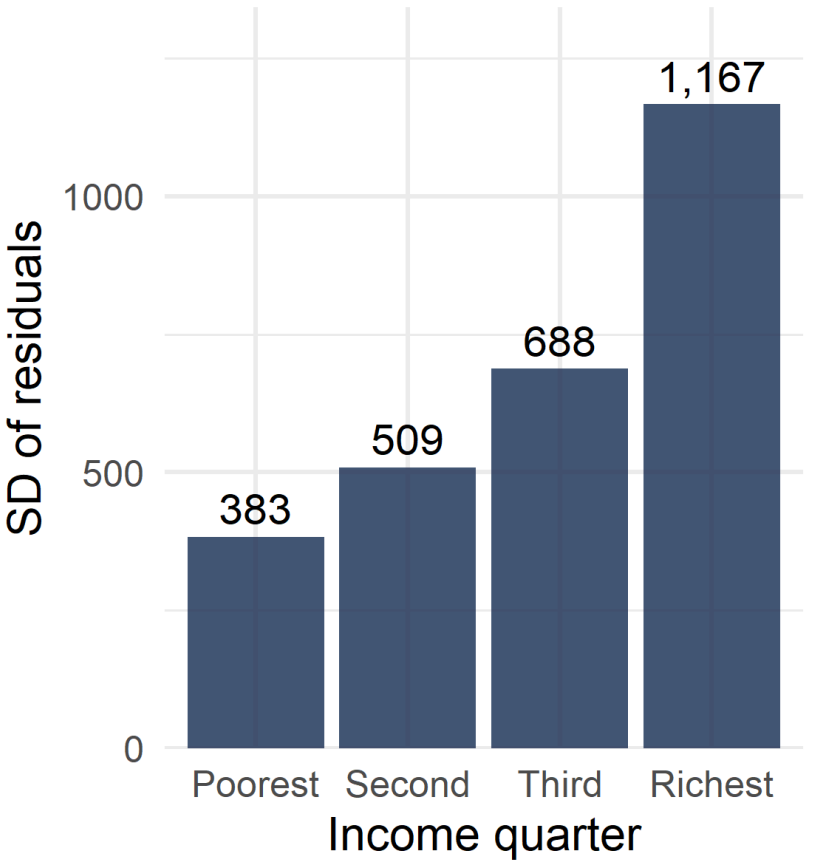
1. Find the 25th, 50th and 75th percentiles of `spending$income_pc`.
2. Compute `sd()` of the residuals for the poorest quarter, the second, the third and the richest quarter.
3. How many times wider is the typical miss for the richest quarter than for the poorest?

Exercise | Start Over | Show Hint | Run Code

```
1
```

The fan





- Poor households all spend about the same on food. Rich households differ a lot.
- One band of $\pm s = \pm 752$ pesos is too wide for the poor and too narrow for the rich.

What a fan breaks — and what it does not

- The slope is still unbiased. Unbiasedness needs only $E(u_i) = 0$, not constant variance.
- The standard-error formula of W08 assumes one common σ^2. With a fan, the reported standard errors are wrong — usually too small.
- So the t-statistics, p-values and confidence intervals are wrong too.
- Prediction intervals have one width for everybody: too wide for the poor, too narrow for the rich.

A manager who reads only the slope sees nothing wrong. The damage is in the precision attached to it.

The Gauss–Markov theorem

Under assumptions 1–4, among all linear and unbiased estimators, OLS has the smallest variance:

$$\mathrm{Var}(\hat{\beta}_1) \leq \mathrm{Var}(\hat{\beta}_1') \qquad \text{and} \qquad \mathrm{Var}(\hat{\beta}_0) \leq \mathrm{Var}(\hat{\beta}_0')$$

OLS is BLUE: the Best Linear Unbiased Estimator.

A linear estimator is a weighted sum of the y_i: $\hat{\beta}_1' = \sum_i c_i y_i$. For OLS the weights are

$$\hat{\beta}_1 = \frac{\sum_i (x_i - \bar{x}) y_i}{\sum_i (x_i - \bar{x})^2} \qquad \text{so} \qquad c_i^{OLS} = \frac{x_i - \bar{x}}{\sum_i (x_i - \bar{x})^2}$$

Research continues: Hansen (2022) showed the word “linear” can be dropped. Under the same assumptions OLS is the best of all unbiased estimators (BUE).

Exam question — Heteroskedastic errors

Question 6 [5 points]:
Consider the linear model of the form:

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i$$

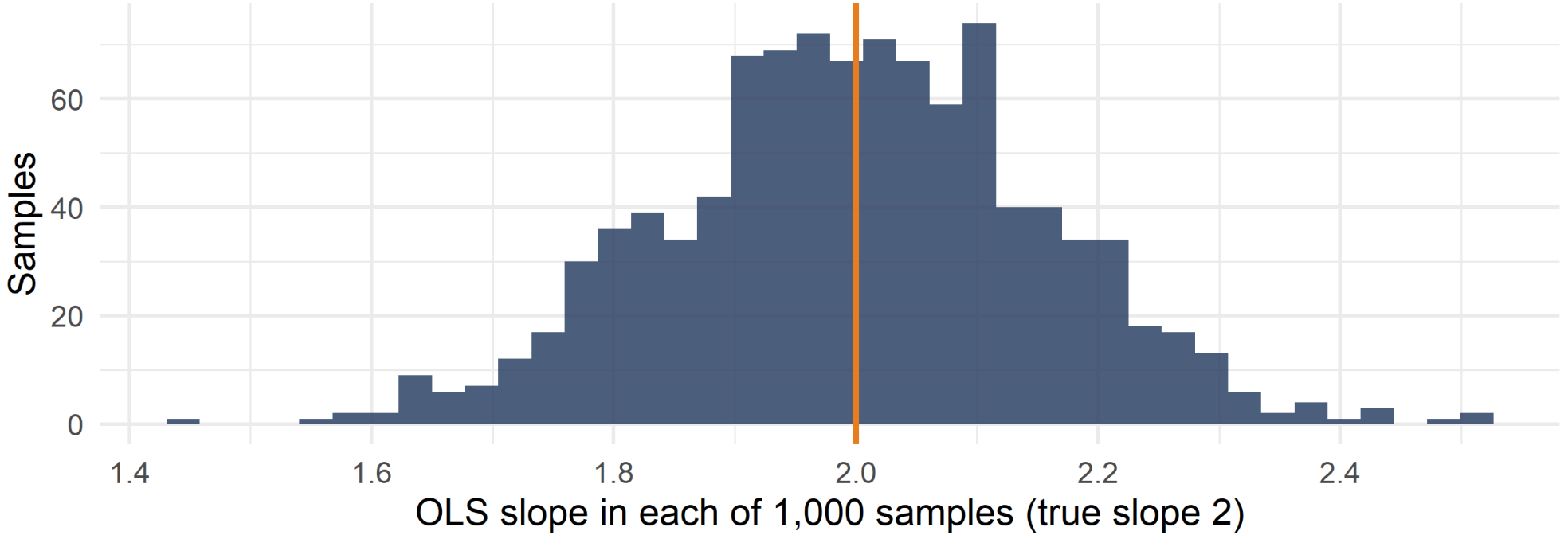
with $E[\epsilon_i] = 0$; $var(\epsilon_i) = \sigma_i^2 \neq \sigma^2$; $cov(\epsilon_i, \epsilon_j) = 0$ for all $i \neq j$, and the estimation of the model by Least Squares. Now consider the following statements:

A: The Least Squares estimators will no longer be unbiased.

B: The Least Squares estimators will no longer have minimum variance.

Then:

Which of the statements A and B are true?



In 1,000 samples where the error spread grows with x, the slopes average 2.000, the true value of 2. So A is false: OLS stays unbiased when the variance is not constant.

B is true. Gauss–Markov needs constant variance (assumption 3) to make OLS the minimum-variance estimator. With $\mathrm{Var}(\epsilon_i) = \sigma_i^2$, an estimator that gives less weight to the noisy observations can beat OLS.

A fix: take logs

`spending` also has the natural logs of both columns:

- `spending$log_food` — log of `spending$food_pc`
- `spending$log_income` — log of `spending$income_pc`

Logs pull in a long right tail. A 1,000-peso difference is large for a poor household and small for a rich one; on the log scale, both are measured as percentages.

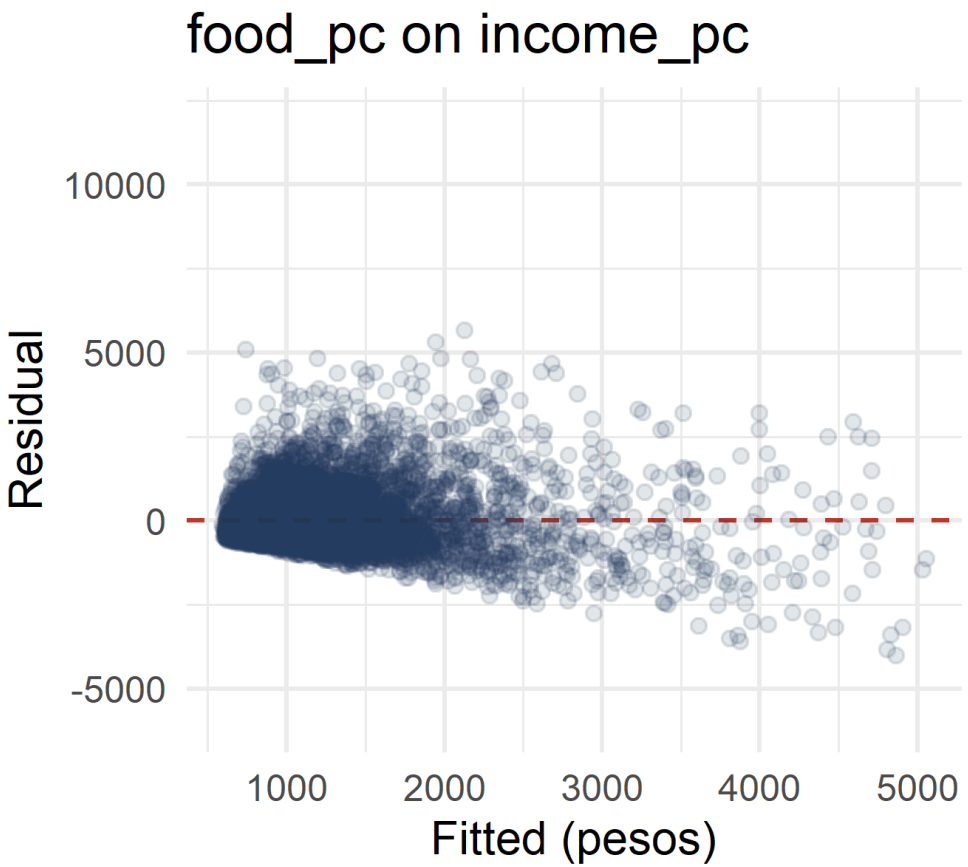
Exercise 12 — Did the logs fix the fan?

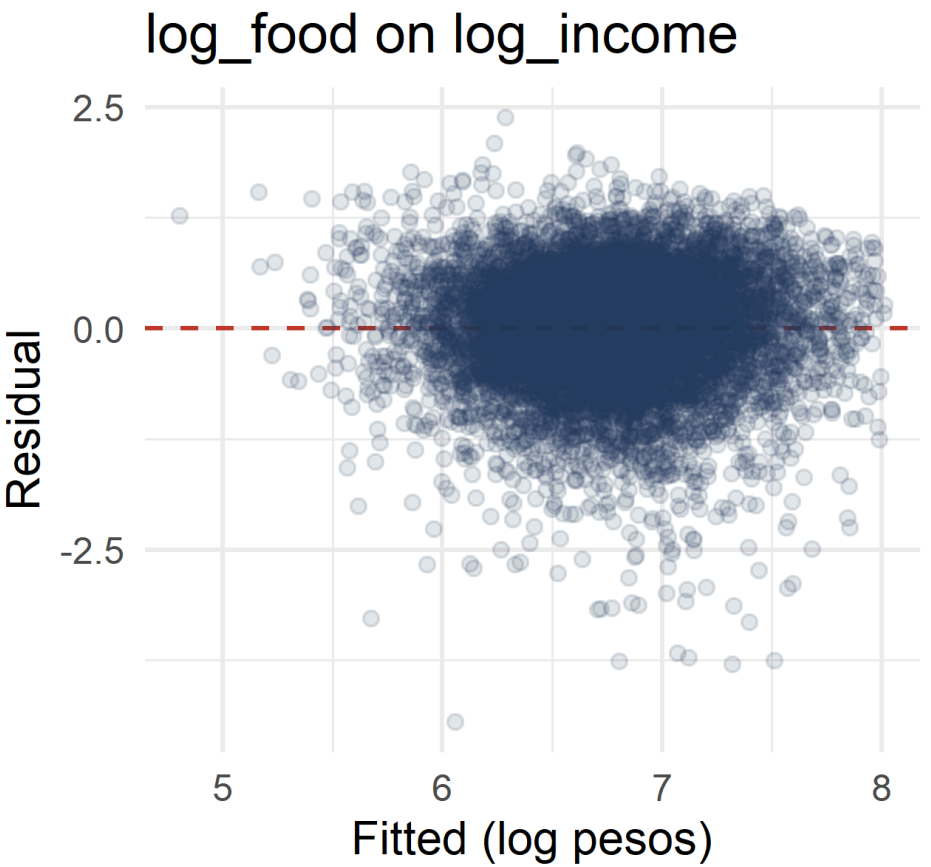
1. Regress `spending$log_food` on `spending$log_income`. Read the slope and R^2.
2. Compute the residual standard deviation in the same four income quarters of `spending$income_pc` as in Exercise 11.
3. Is the spread still growing?

Exercise | Start Over | Show Hint | Run Code

```
1
```

Levels against logs





- Left: a fan. Right: an even band.
- The R^2 values are not directly comparable: the two models explain different y variables.

Reading a slope after logs

Model	Meaning of $\hat{\beta}_1$
y on x	one more unit of x goes with $\hat{\beta}_1$ more units of y
$\log(y)$ on x	one more unit of x goes with about $100\ \hat{\beta}_1$ percent more y
$\log(y)$ on $\log(x)$	one percent more x goes with about $\hat{\beta}_1$ percent more y — an elasticity

Food: $\hat{\beta}_1 = 0.518$. A household with 10% more income per person spends about 5.2% more on food per person. Food spending rises with income, but less than in proportion — food is a necessity.

The other fix keeps the model and corrects only the standard errors (heteroskedasticity-robust standard errors). It comes later in the course.

6 · Are the errors independent?

Assumption 4: one observation’s error says nothing about the next one’s.

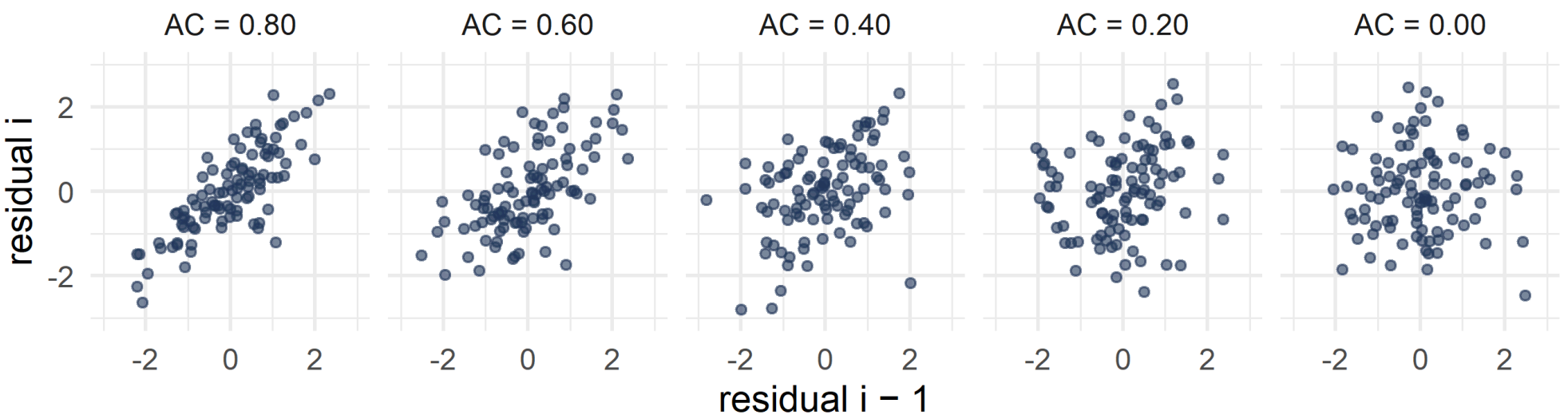
Errors that remember

When the data come in time order, check whether each residual looks like the one before:

1. Plot each residual $\hat{u}_i$ against the previous one, $\hat{u}_{i-1}$: $\hat{u}_2$ against $\hat{u}_1$, $\hat{u}_5$ against $\hat{u}_4$, and so on.
2. Compute their correlation. This is the lag-1 autocorrelation.
3. Do the same with longer gaps: $\hat{u}_i$ against $\hat{u}_{i-2}$, $\hat{u}_{i-3}$, … Note the correlation at each lag: $\rho(\hat{u}_i, \hat{u}_{i-1})$, $\rho(\hat{u}_i, \hat{u}_{i-2})$, $\rho(\hat{u}_i, \hat{u}_{i-3})$.

If the errors are uncorrelated, every one of these correlations should be close to 0.

What autocorrelation looks like



From left to right the tilt fades. At the right, knowing yesterday's residual tells you nothing about today's.

Ecobici days come in order

`bikes$date` runs from January 2017 to December 2019. Rain, a metro closure or a holiday week lasts more than one day.

If temperature misses a day because of a metro closure, will it miss the next day too?

✏️ Exercise 13 — Does today's miss predict tomorrow's?

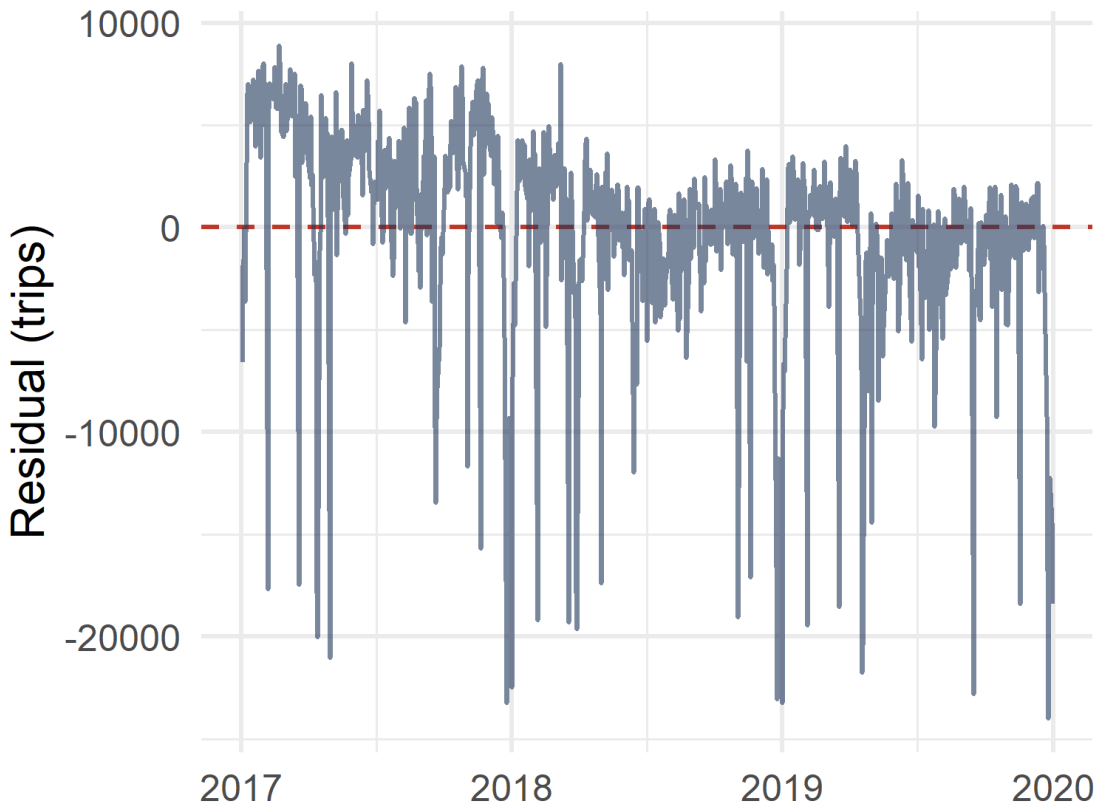
Use `bikes` (`bikes$date`, `bikes$trips`, `bikes$temp`; 781 weekdays in date order).

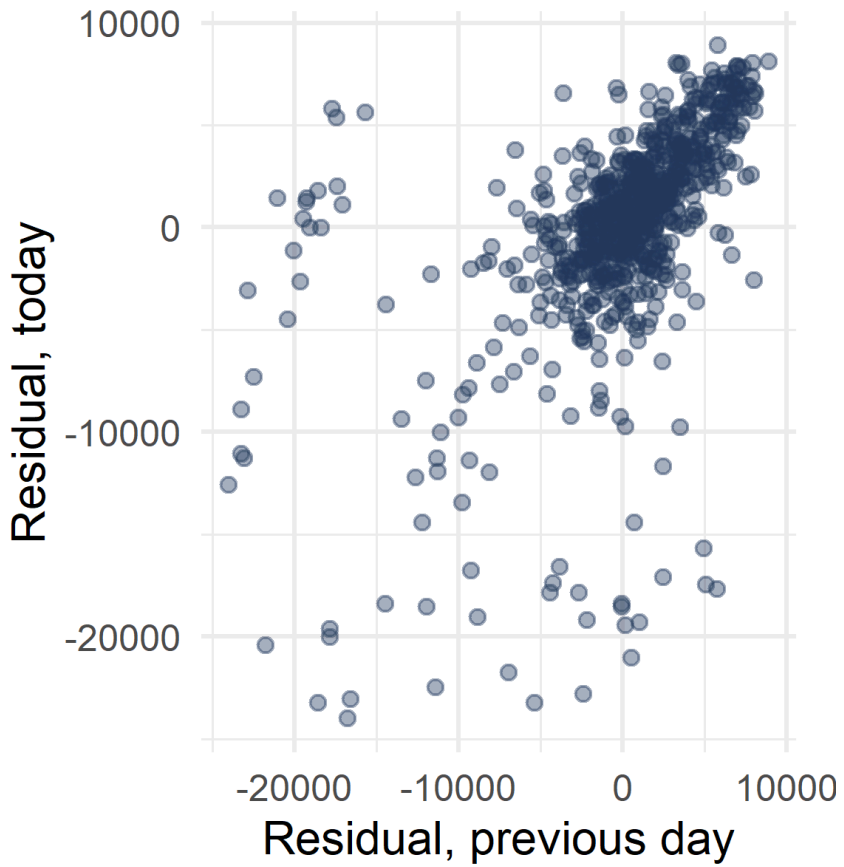
1. Plot the residuals of the regression of `bikes$trips` on `bikes$temp` against `bikes$date`. Do you see long runs above or below zero?
2. Compute the correlation between each day's residual and the previous day's (lag 1).
3. Do the same with a gap of two days (lag 2).

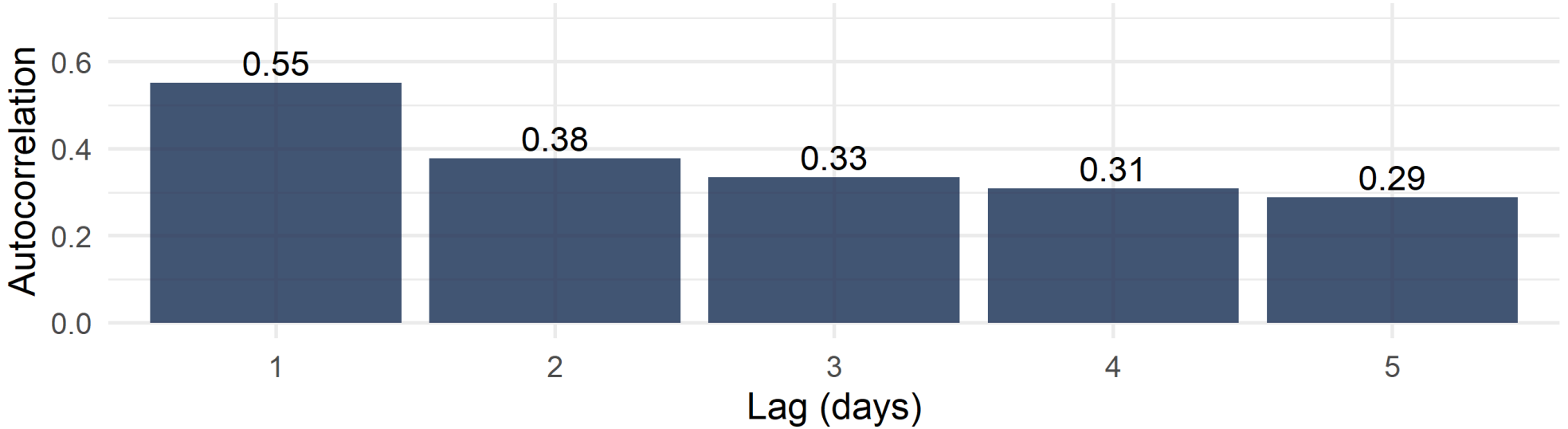
Exercise | Start Over | Show Hint | ▷ Run Code

```
1 reg <- lm(trips ~ temp, data = bikes)
```

Ecobici residuals over time







- The residuals move in waves: weeks above the line, then weeks below it.
- The autocorrelation is 0.55 at lag 1 and still 0.29 at lag 5.

What correlated errors break

- The slope is still unbiased if $E(u_i) = 0$.
- But 781 correlated days carry less information than 781 independent days. The standard errors from `summary()` treat every day as new information, so they are too small.
- The t-statistics are too large and the p-values too small. Results look more certain than they are.

For the Ecobici client, the slope of 724 trips per degree is a fair estimate. The precision printed next to it is too optimistic.

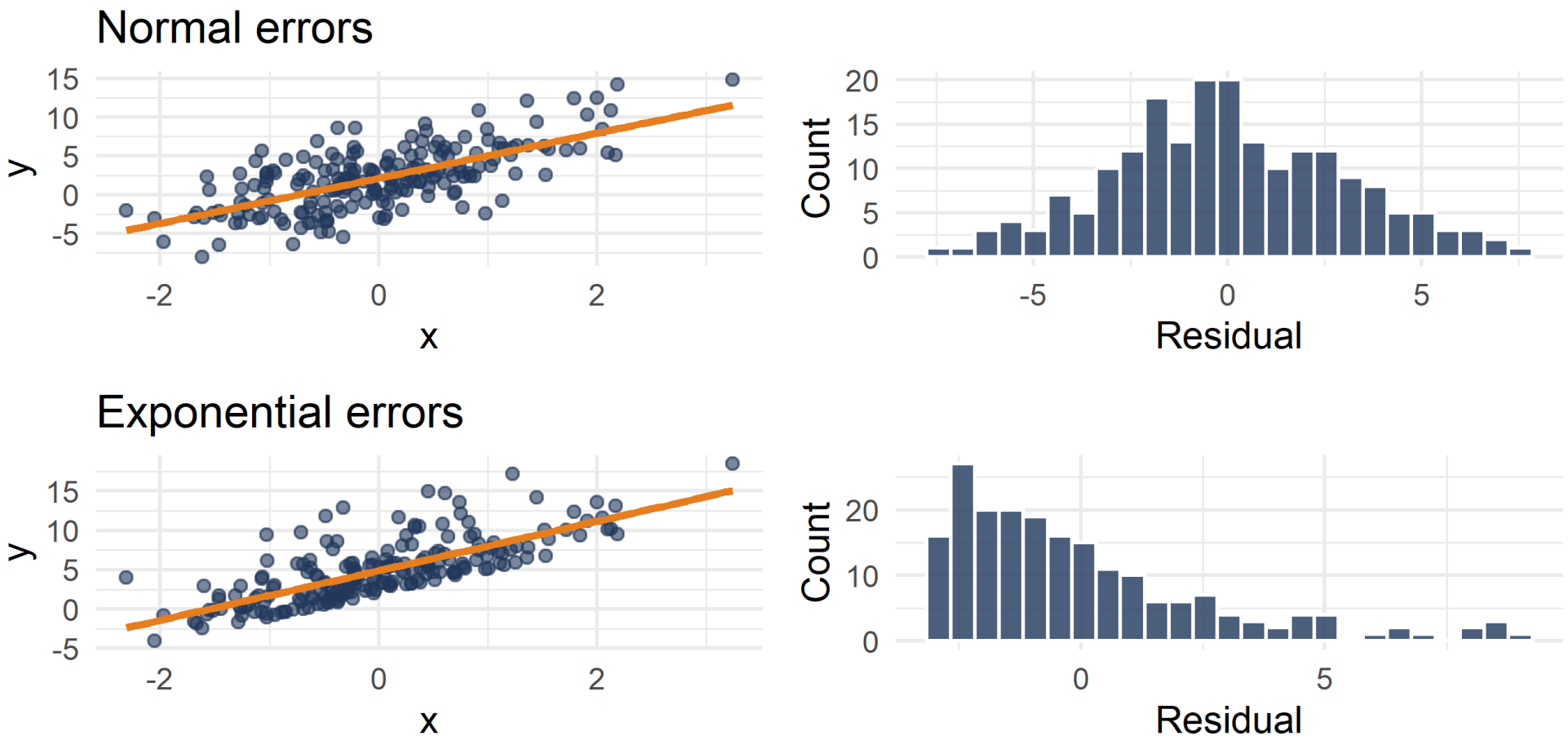
7 · Are the errors normal?

Assumption 5. Pictures first, then a formal test.

Why normality matters

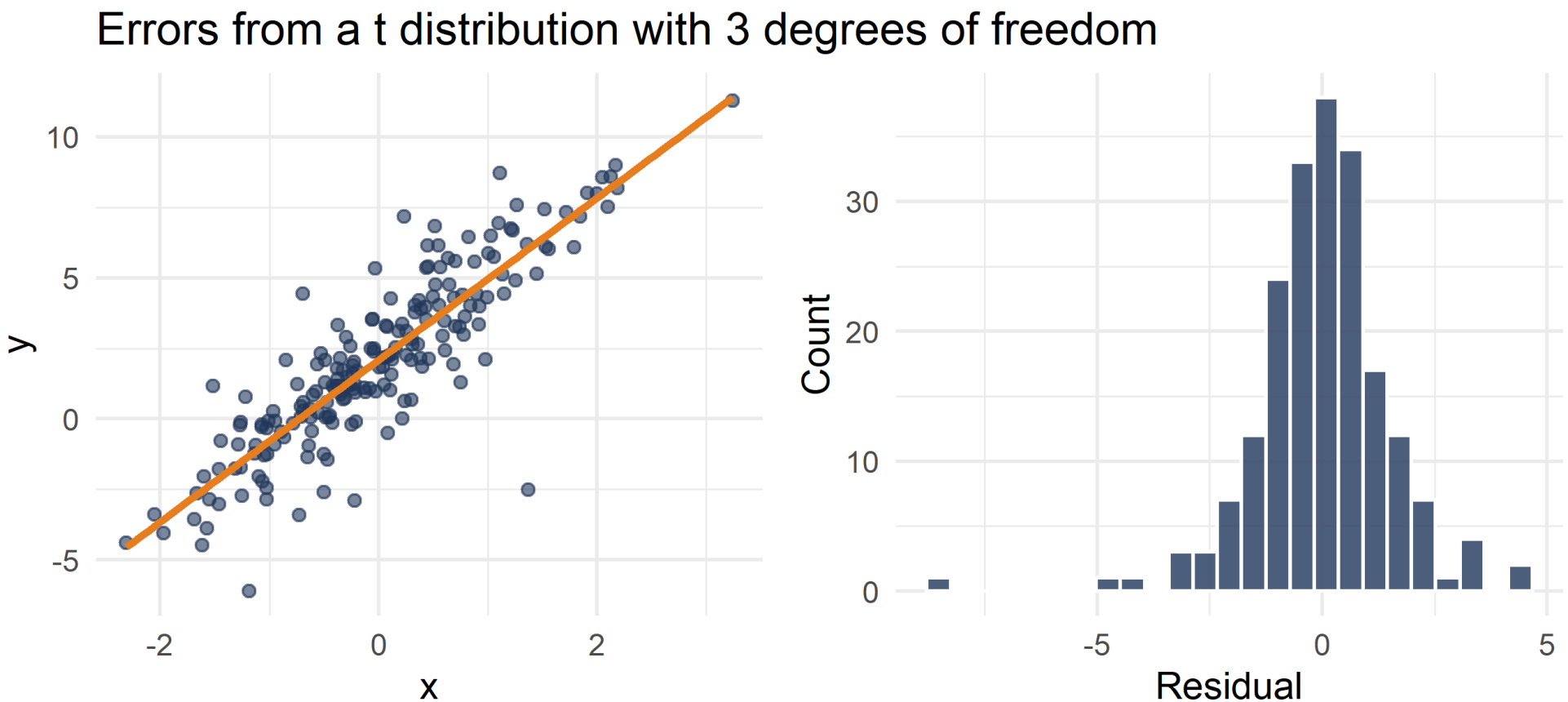
- The slope, the intercept and R^2 do not need normal errors. OLS computes them either way.
- The t-test, the F-test and the intervals of W08 use the t and F distributions. Those are exact only when the errors are normal.
- With a large sample, the central limit theorem makes $\hat{\beta}_1$ close to normal anyway, so the tests still work approximately. With a small sample there is no such protection.

Histograms: some departures are obvious



Top: a bell. Bottom: a long right tail — clearly not normal.

Histograms: some are not obvious



This looks almost like a bell. But the errors come from a t_3 distribution: too many values far from the centre. A histogram cannot show this well. We need a sharper tool.

The Q-Q plot

A quantile–quantile plot compares the quantiles of the residuals with the quantiles they would have if they came from a normal distribution.

1. Standardise the residuals: subtract the mean, divide by the standard deviation.
2. Sort them from smallest to largest. Give each an index i: 1 is the smallest, n the largest.
3. Each one's cumulative probability is the share of the data below it:

$$CP_i = \frac{i}{n+1}$$

4. Find the standard normal quantile with that probability, `qnorm(CP)`.
5. Plot the sample values against the normal quantiles. If the residuals are normal, the points lie on the 45° line.

Exercise 14 — Build a Q-Q plot by hand

Use the landlord's regression of `rents$rent` on `rents$size` (150 apartments).

1. Standardise the residuals: call the result `z`.
2. Make the cumulative probabilities `p` for $i = 1, \ldots, 150$.
3. Plot the sample quantiles of `z` against the normal quantiles `qnorm(p)`.
4. Do the points follow the 45° line?

Exercise | Start Over | Show Hint | Run Code

```
1
```

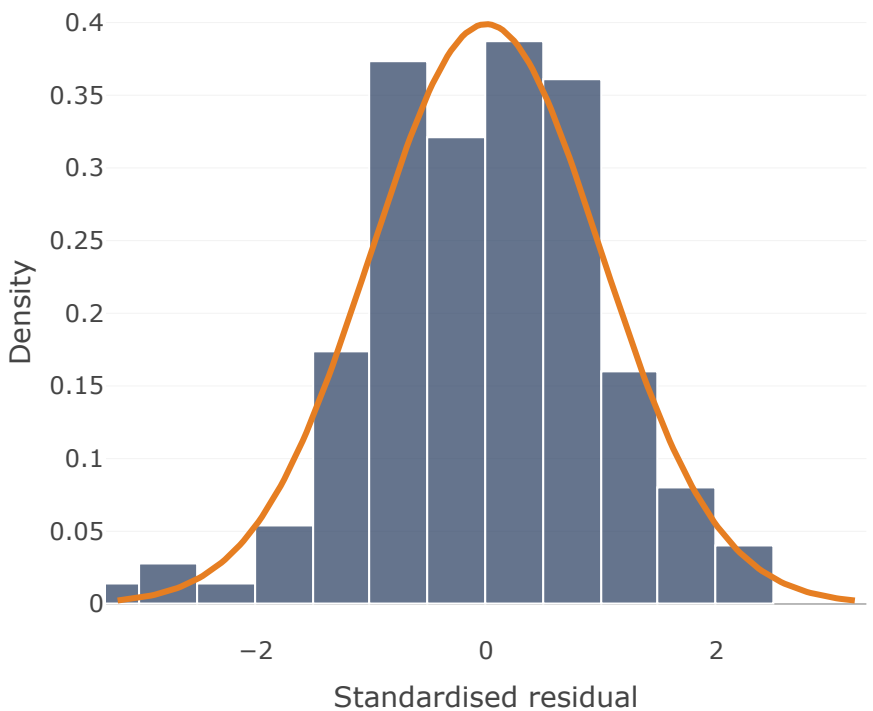
The Q-Q table

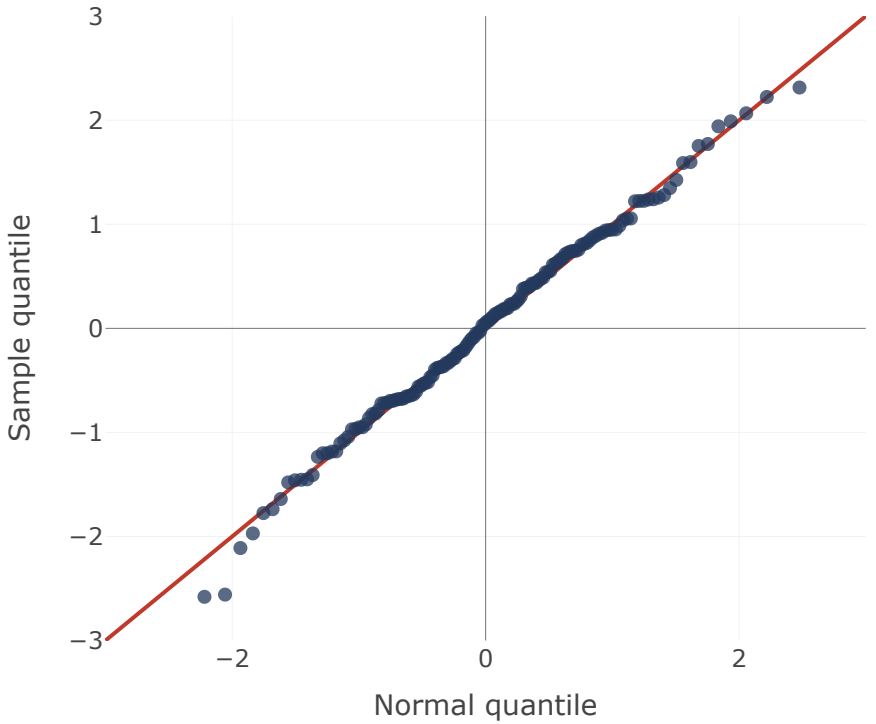
Index i	CP = i/151	Normal quantile	Standardised residual
1	0.007	-2.477	-3.110
2	0.013	-2.219	-2.580
3	0.020	-2.056	-2.558
4	0.026	-1.935	-2.111
74	0.490	-0.025	0.029
75	0.497	-0.008	0.040
76	0.503	0.008	0.064
77	0.510	0.025	0.073
147	0.974	1.935	1.990
148	0.980	2.056	2.067
149	0.987	2.219	2.224
150	0.993	2.477	2.315

The smallest four, the middle four and the largest four of the 150 rows. In each row the last two columns are close: the residuals sit where a normal distribution would put them.

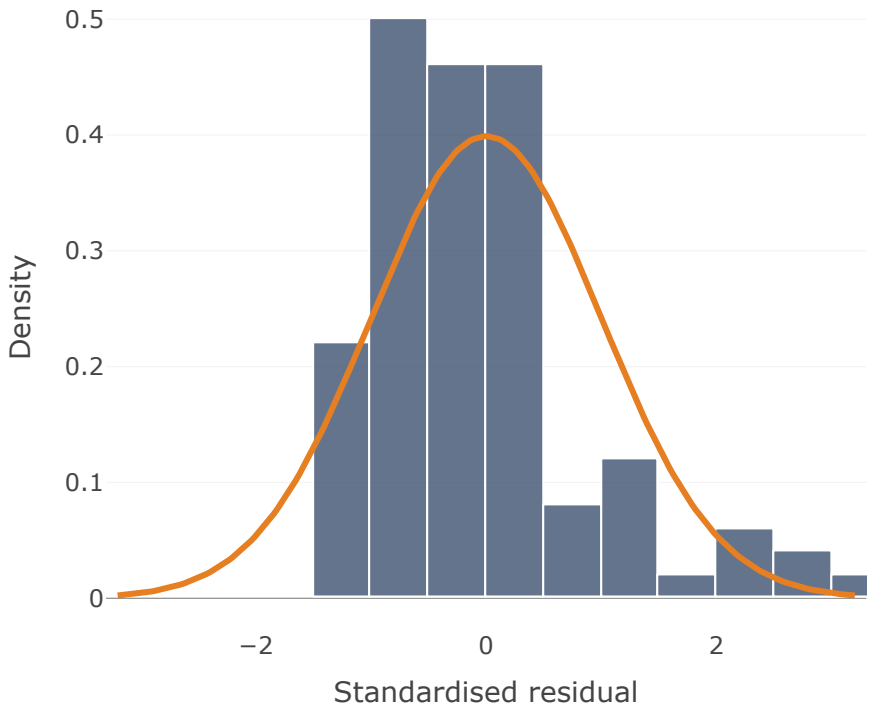
The landlord's residuals, both ways

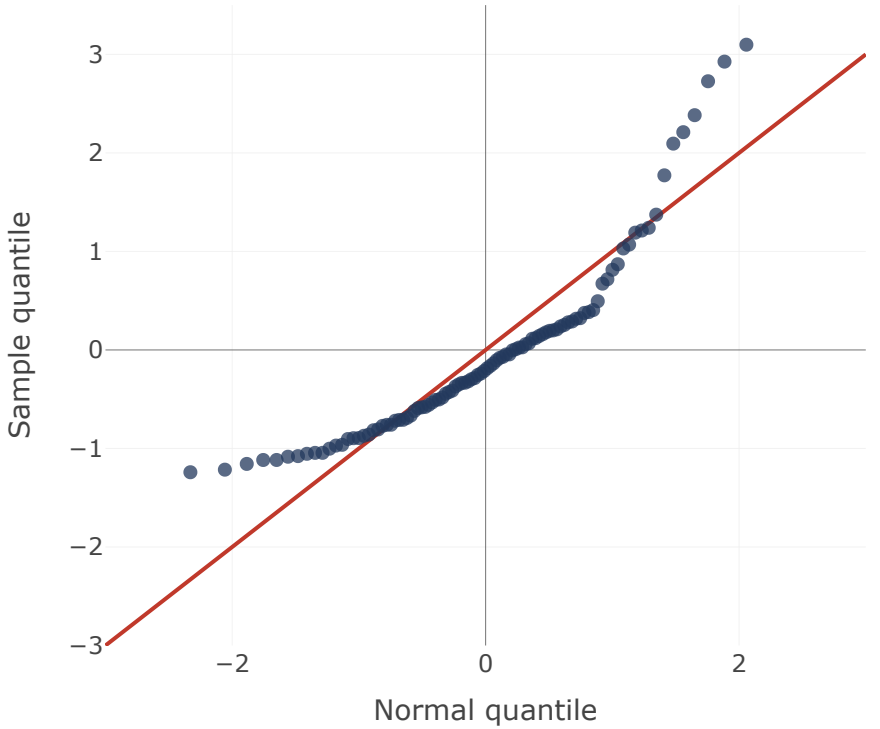
Hover over the points.





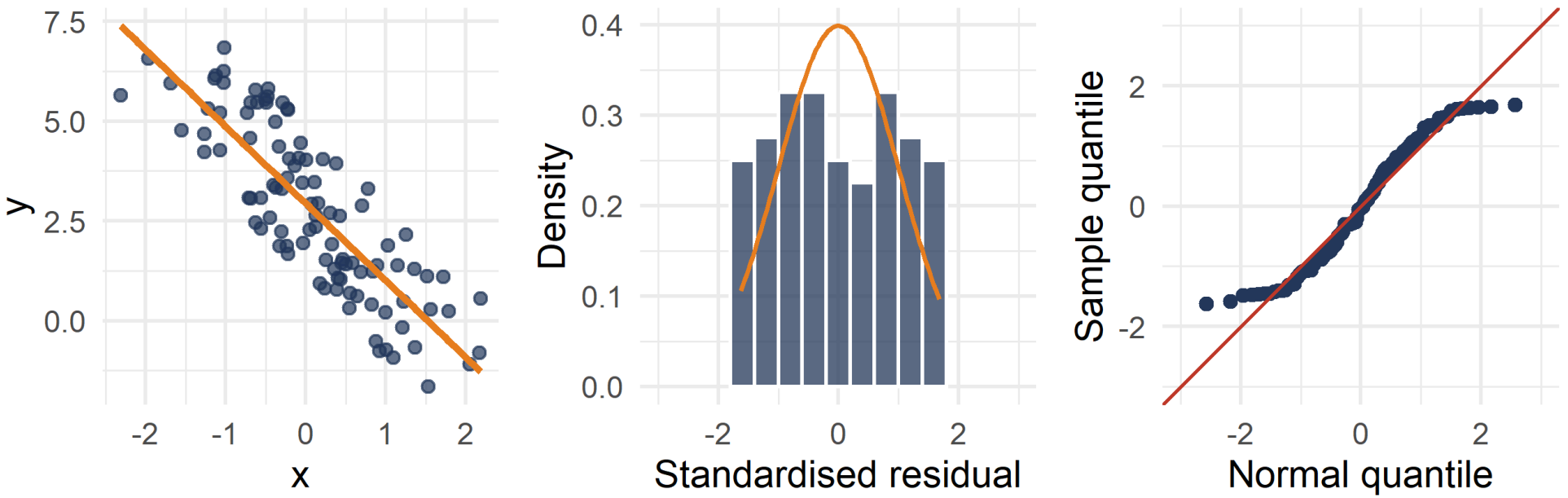
Q-Q plot: skewed residuals





Residuals from a chi-square distribution. The points bend away from the line into a banana: the right tail is too long, the left tail too short.

Q-Q plot: residuals with light tails



Uniform errors: no values far out. The Q-Q plot makes an S — the smallest residuals are not small enough and the largest not large enough.

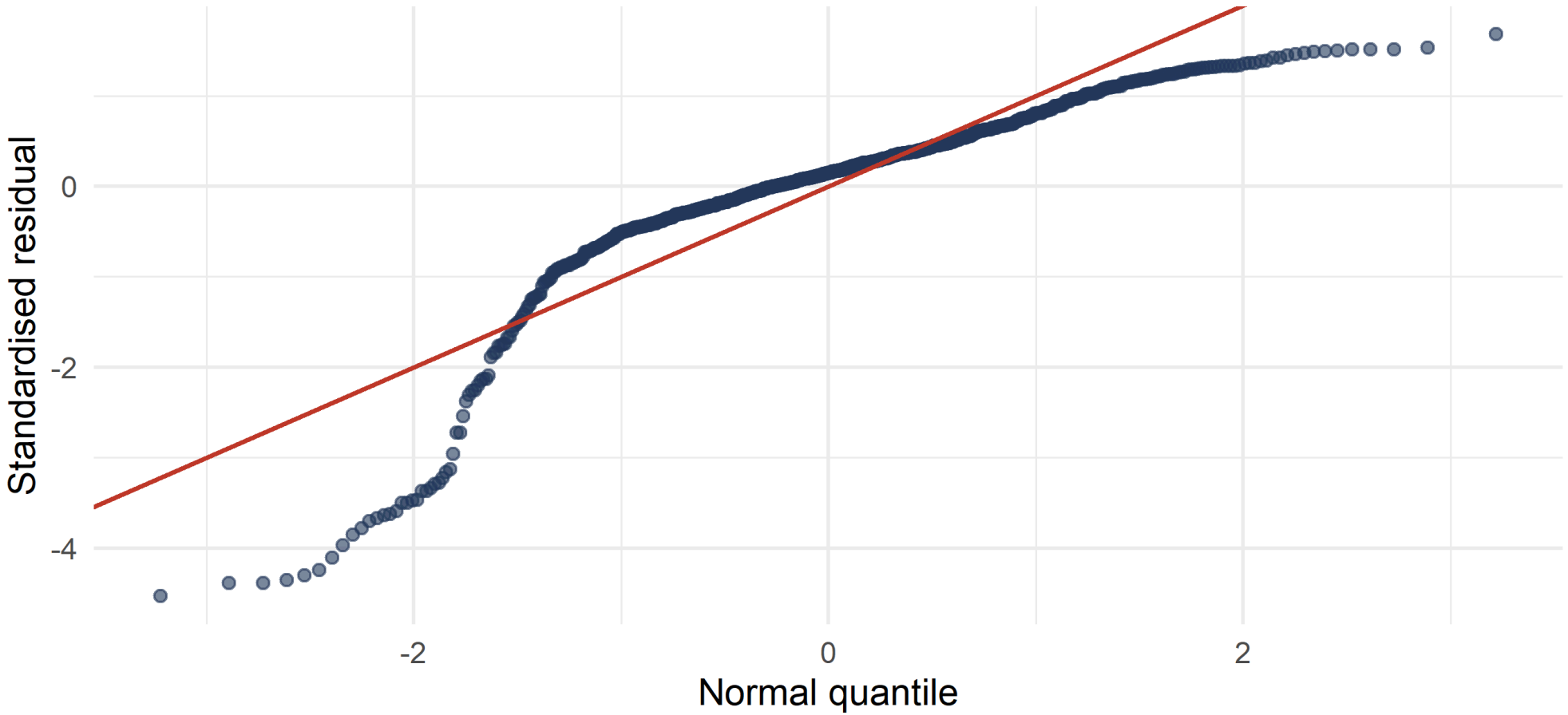
Exercise 15 — The Ecobici Q-Q plot

Repeat Exercise 14 for the regression of `bikes$trips` on `bikes$temp` (781 days). Which tail leaves the line? What kind of days are they?

Exercise | Start Over | Show Hint | Run Code

```
1 reg <- lm(trips ~ temp, data = bikes)
```

The Ecobici Q-Q plot

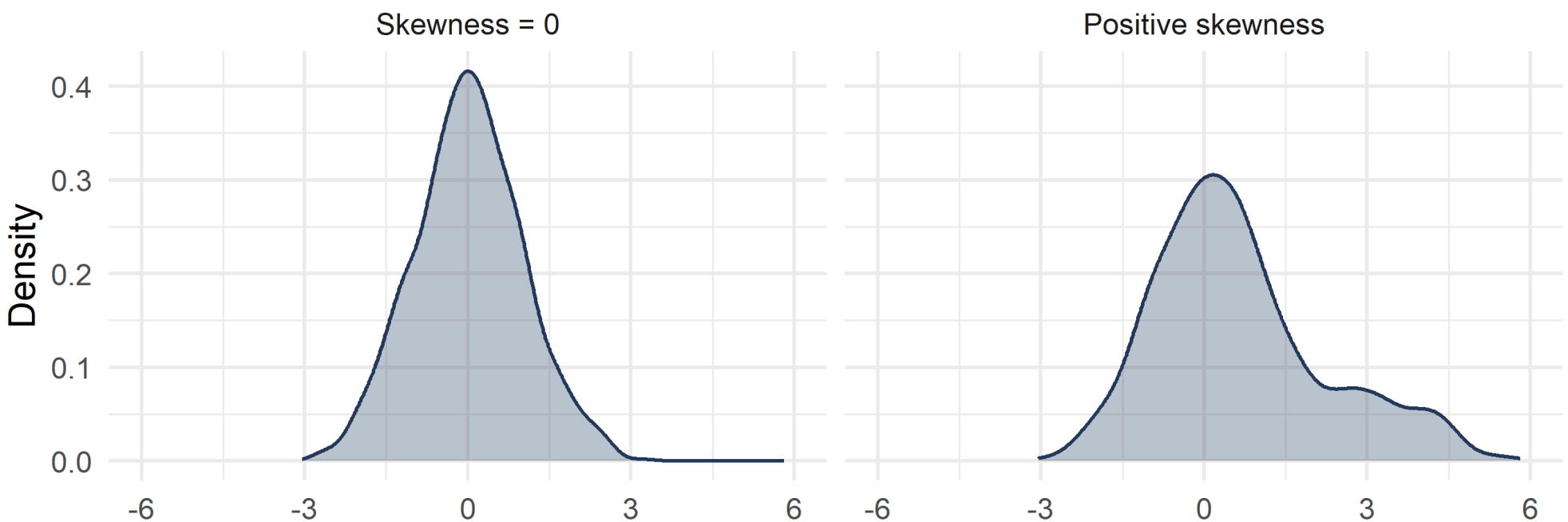


- Long left tail: the holidays.
- A picture says “not normal”. A test says how sure we can be.

Jarque–Bera: skewness

A normal distribution is symmetric: its skewness is 0.

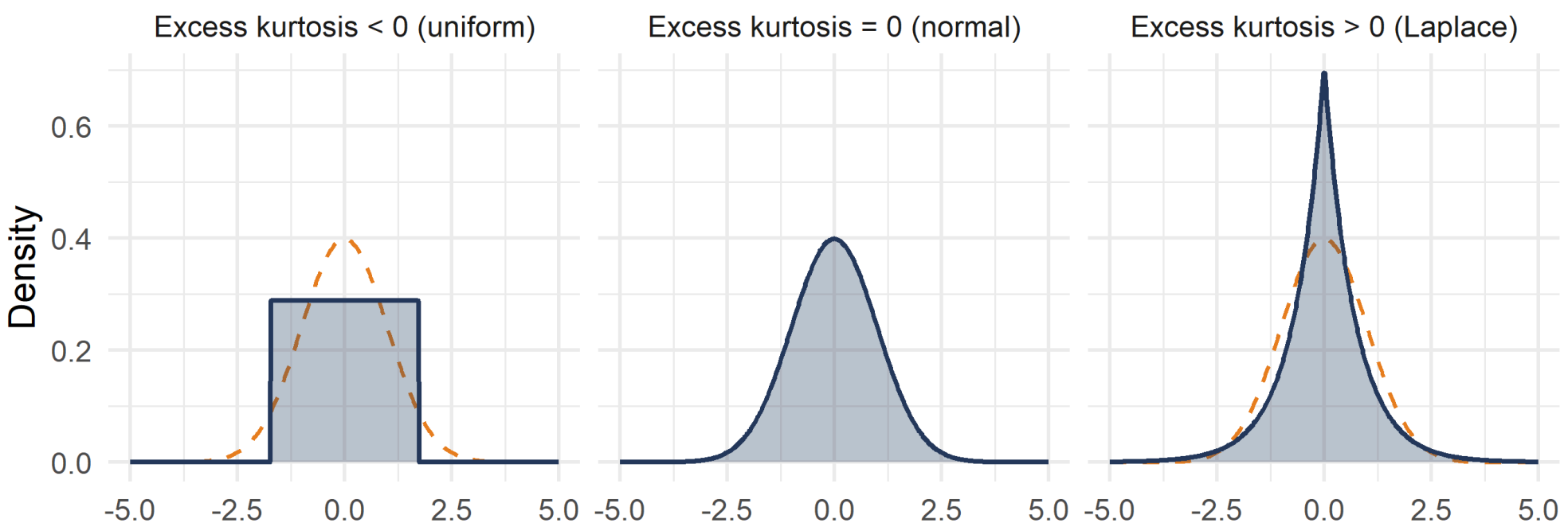
$$S = \frac{\frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^3}{\left(\frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2\right)^{3/2}}$$



Jarque–Bera: kurtosis

The kurtosis of a normal distribution is 3. Excess kurtosis subtracts that 3, so for a normal distribution it is 0:

$$EK = \frac{\frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^4}{\left(\frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2\right)^2} - 3$$



All three have variance 1. The dashed orange curve is the normal, for reference.

The Jarque–Bera statistic

$$JB = \frac{n}{6}\left(S^2 + \frac{EK^2}{4}\right)$$

- JB is large when the skewness is far from 0 or the excess kurtosis is far from 0.
- If the data come from a normal distribution, JB follows a chi-square distribution with 2 degrees of freedom — one for skewness, one for kurtosis.

H_0: the residuals come from a normal distribution (JB is small) against H_A: they do not (JB is large).

Only a large JB is evidence against H_0, so the test is one-sided. Reject at level α if $JB > \chi^2_{1-\alpha;\,2}$. The p-value is $P(\chi^2_2 \geq JB)$.

Exercise 16 — Test the landlord's residuals

Use the regression of `rents$rent` on `rents$size` ($n = 150$). At the 5% level, test

H_0: the residuals are normal against H_A: they are not.

1. Compute the skewness S and the excess kurtosis EK of the residuals.
2. Compute JB, the 5% critical value and the p-value.
3. Reject or not?

Exercise | Start Over | Show Hint | Run Code

```
1
```

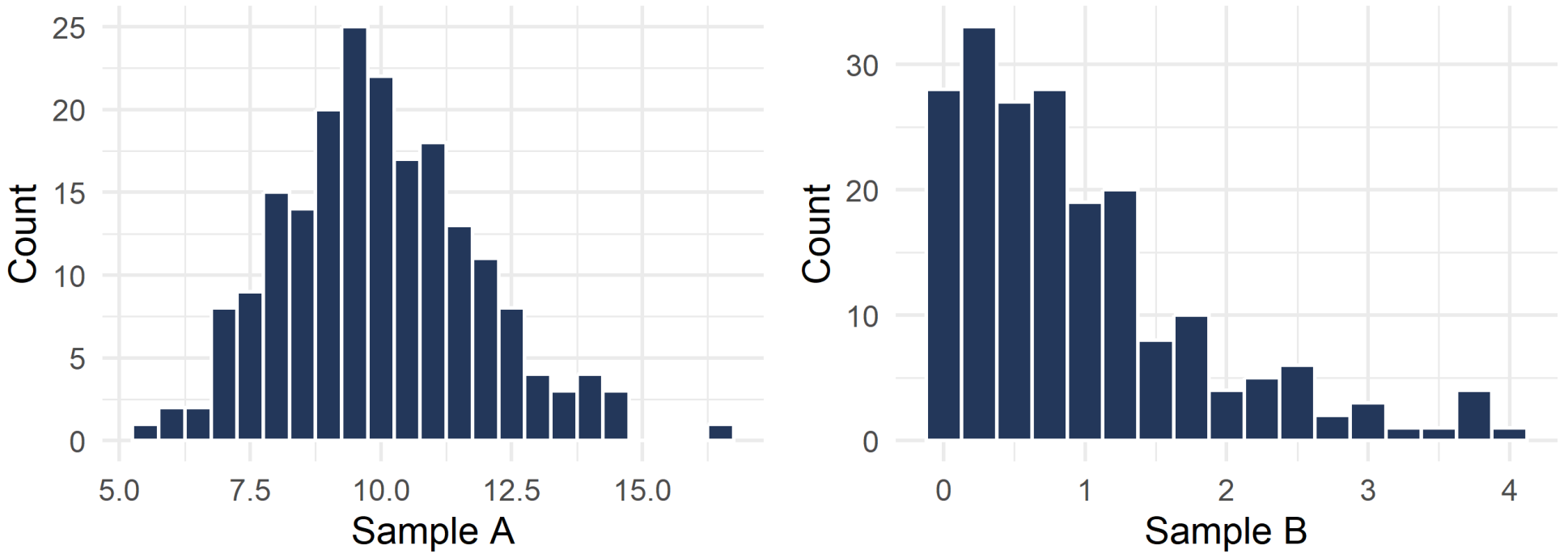
Exercise 17 — Test the Ecobici residuals

Same test for the regression of `bikes$trips` on `bikes$temp` ($n = 781$): H_0: the residuals are normal against H_A: they are not, at the 5% level.

Exercise | Start Over | Show Solution | Run Code

```
1
```

Two samples, two verdicts



Two samples of 200. Which one would the Jarque–Bera test reject?

Sample A (drawn from a normal): $S = 0.351$, $EK = 0.167$, $JB = 4.33$, p-value 0.114. Do not reject.

Sample B (drawn from an exponential): $S = 1.400$, $EK = 1.692$, $JB = 89.1$, p-value below 0.001. Reject: not normal.

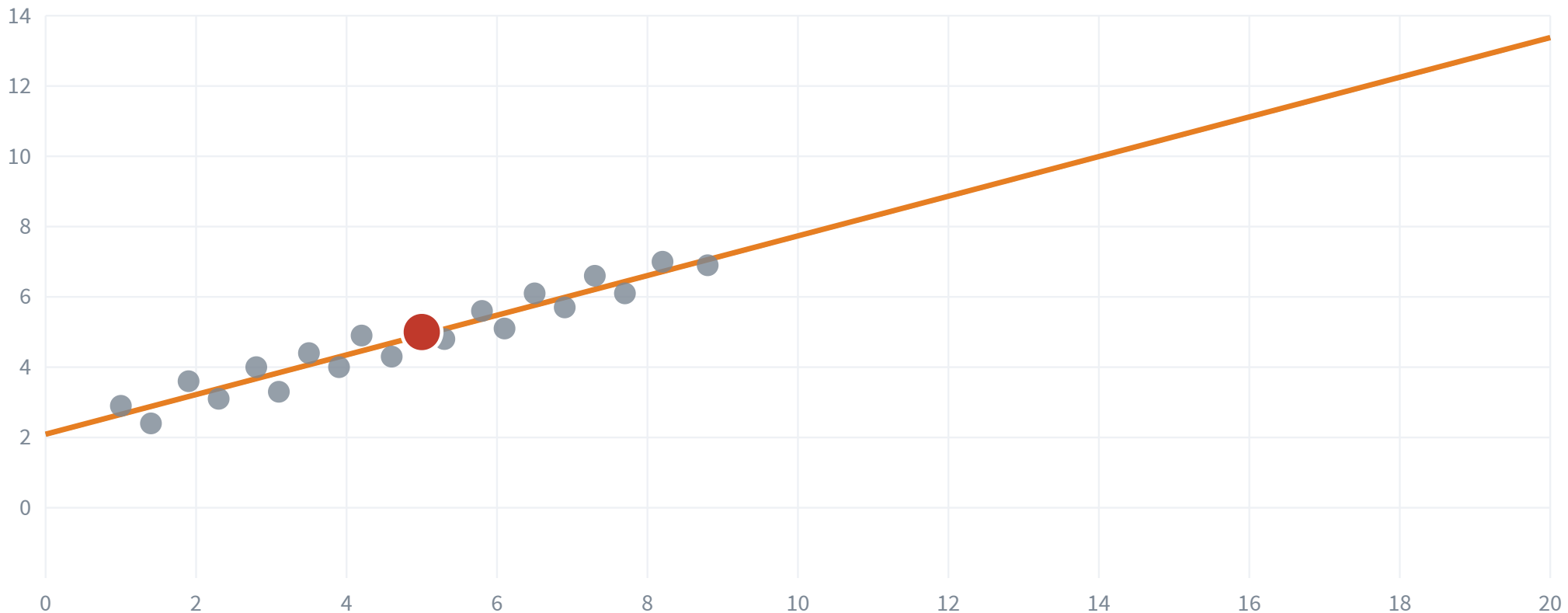
8 · One point can move the line

Not an assumption — a warning. Some observations have far more say in the slope than others.

Drag one point

Grey points are fixed. Drag the red point. Where does one point pull the line the most: in the middle of the x range, or far out to one side?

Put it back | Show the slopes



slope without the red point ? slope with it ?

Leverage

- A point in the middle of the x range, moved up or down, mostly shifts the line up or down. The slope barely changes.
- A point far from $\bar{x}$ acts like a long lever. Moved up or down, it rotates the line.
- A point far from the others in x has high leverage. If it also sits off the line, it is influential: one observation decides the slope.

An influential point is not an assumption failure. It is a warning that the estimate is fragile. The check: fit the model with and without it, and compare.

Cold days, empty streets

In W07 the Ecobici residual histogram had a long left tail. Section 7 found the same tail in the Q-Q plot. Those days — Christmas, New Year's Day, Holy Week — are also some of the coldest days in the data.

Do a few holidays change the answer to the client's question, the trips per degree?

Exercise 18 — Take out the holidays

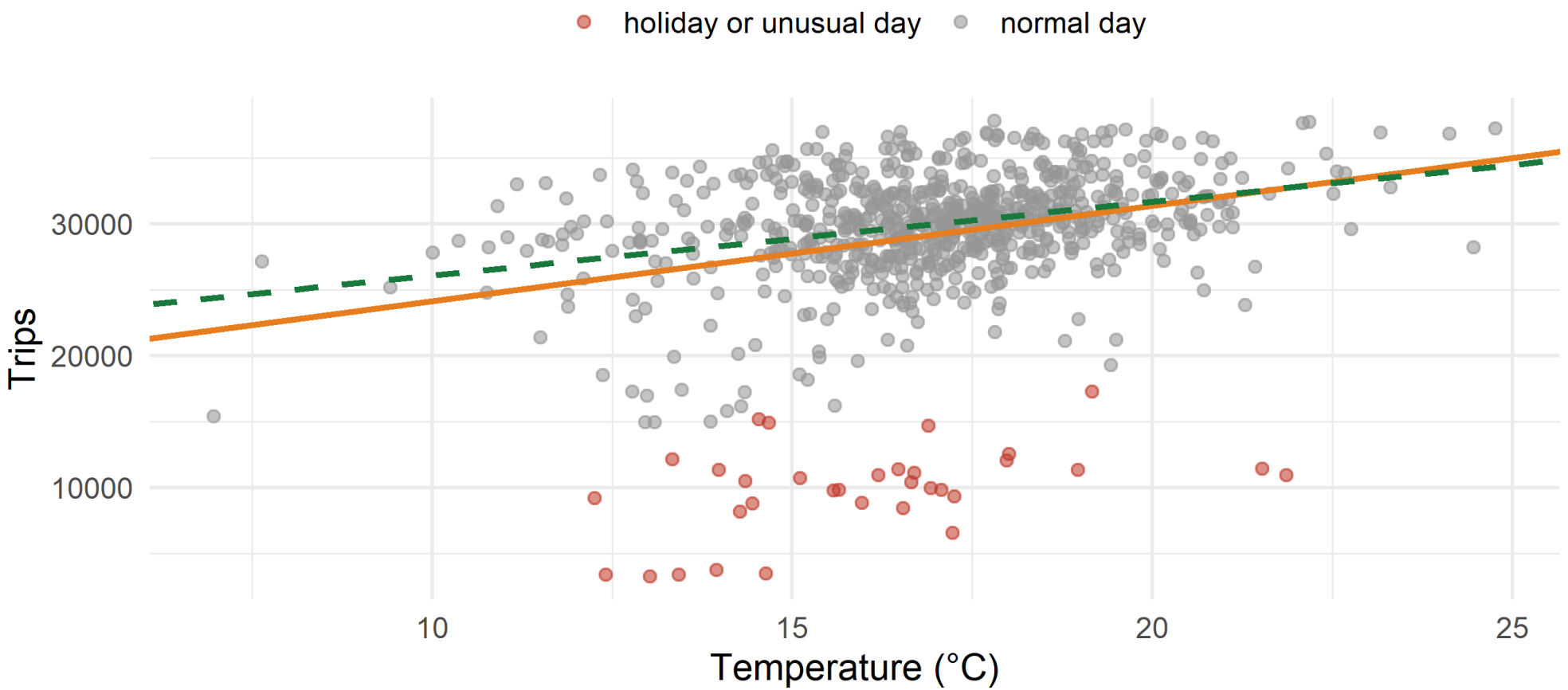
Use `bikes` (`bikes$trips`, `bikes$temp`, `bikes$date`).

1. Regress `bikes$trips` on `bikes$temp`. Keep only the days whose residual is above −12,000 trips, in a new data frame `bikes_normal`.
2. How many days did you drop? Print their dates.
3. Refit the regression on `bikes_normal`. Compare the slope and R^2 with 723.6 and 0.088.

Exercise | Start Over | Show Hint | Run Code

```
1
```

Holidays tilt the line



- Orange: all 781 days, slope 723.6. Dashed green: without the 33 red days, slope 559.4.
- The red days sit low and mostly on the cold side, so they pull the left end of the line down and make it steeper.

For the client: the fair number for an ordinary weekday is closer to 560 trips per degree. Holidays belong in the model as their own variable — that is multiple regression, from W10.

9 · Putting it together

One checklist, one verdict, and one question the residuals cannot answer.

The diagnostic checklist

Assumption	Check	If it fails	What to do
linearity	residuals against fitted: curve or arch?	coefficients describe the wrong shape; forecasts are wrong in a pattern	transform x or y
constant variance	residuals against fitted: fan? SD by group	slope fine; standard errors, tests, intervals wrong	logs; robust standard errors
uncorrelated errors	residuals in time order; lag correlations	slope fine; standard errors too small	model the time pattern
normality	histogram, Q-Q plot, Jarque–Bera	small samples: tests and intervals unreliable	large n helps; transform y
no single point in charge	refit without unusual points	one observation decides the slope	report both; find out why

Always fit first, then check. The residual plots are the first thing to look at.

The Ecobici verdict

Fit: $R^2 = 0.088$ and a typical miss of 5,302 trips. But $F = 75.3$: the temperature effect is real.

Diagnostics: the line is the right shape. But the residuals are strongly autocorrelated (0.55), not normal (JB = 1,390), and their spread is larger on cold days, which is where the holidays fall.

So the slope is usable, but the standard errors from `summary()` are too optimistic, and 33 holidays pull the slope up by about 160 trips per degree. The model is not useless. It is incomplete.

Is the slope itself right?

A firm regresses employees' salary on their years of education to set its pay scale.

- What kinds of factors end up in the error u_i?
- Are they correlated with education?
- Would the slope estimator be unbiased?

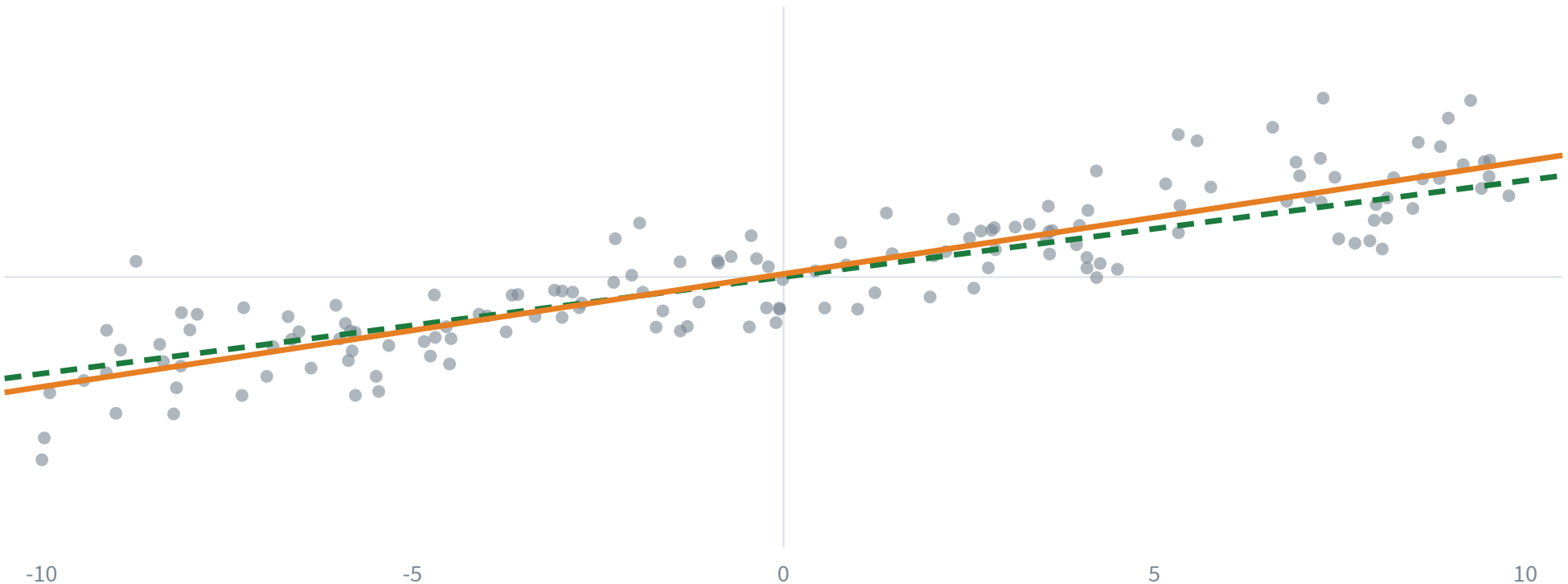
Ability, family background, connections and ambition all sit in u_i, and all of them move with education. Then $E(u_i) = 0$ fails at a given level of education: the error is higher, on average, for the more educated. The slope picks up part of their effect and is biased — usually upwards.

No residual plot can show this. OLS makes the residuals uncorrelated with x by construction, even when the true errors are not.

When the error moves with x

The dashed green line is the truth: $y = 0.5x$. Grey points are the data, the orange line is the OLS fit.

Mean of the error 0.0 Error moves with x 0.00 New sample



true line: intercept 0.00, slope 0.50 | OLS: intercept 0.16, slope 0.58

- Both sliders at 0: the assumptions hold, and OLS lands on the truth, up to noise.
- Move the mean of the error: the whole cloud shifts, and the intercept absorbs it.
- Make the error move with x: the slope is biased, and more data does not fix it.

✏️ Exercise 19 — Can Finance trust the salary model?

A tech company in Guadalajara budgets hiring with a regression of salary on experience. The data frame `hires` has its 120 engineers:

- `hires$experience` — years of experience
- `hires$salary` — monthly gross salary, in pesos

Give Finance the full check of the regression of `hires$salary` on `hires$experience`:

1. Fit: R^2 and the typical miss s, in pesos.
2. Test $H_0 : \beta_1 = 0$ against $H_A : \beta_1 \neq 0$ with the F statistic, at 5%.
3. Linearity and constant variance: the residual plot, and the residual standard deviation in each quarter of `hires$experience`.
4. Normality: test H_0: the residuals are normal against H_A: they are not, with Jarque–Bera at 5%.
5. One sentence for Finance: can they use the slope and the p-values in `summary()`?

Exercise | Start Over | Show Solution | Run Code

```
1
```

Exam question — Spring 2026 Midterm 2, Q25–Q26

Real questions from last spring's Midterm 2. A FinTech app, $n = 500$ users. The data frame `fintech` is loaded below:

- `fintech$MonthlyVolume` — the user's monthly transaction volume
- `fintech$LinkedAccounts` — how many bank accounts the user has linked to the app

Q25. Regress MonthlyVolume on LinkedAccounts. Calculate SST, SSR, SSE. Report SSR.

Q26a. Using the F-test, test whether LinkedAccounts explains any variation in MonthlyVolume at the 1% significance level. State your hypotheses. Calculate the F-statistic.

Q26b. Find the critical value for this F-test and state your conclusion. Do you reject or fail to reject?

Exercise | Start Over | Show Hint | Run Code

```
1
```

Exam question — Spring 2026 Midterm 2, Q30

Q30. The exam showed a plot of the residuals from a linear regression against the predictor. Comment on what you observe in this plot. Does the plot suggest that all OLS assumptions are satisfied? If not, which assumption might be violated, and what would be the consequences?

The data behind that plot are in `q17`: the predictor `q17$x` and the outcome `q17$y`. Draw the plot yourself, then answer.

Exercise | Start Over | Show Hint | Run Code

```
1
```

→ More exam practice at the end of this deck

What you should be able to do

- Compute the residual standard error s and say what it means in the units of y.
- Build the ANOVA table, the F statistic, its critical value and p-value, with both hypotheses stated.
- Show that the correlation test, the t-test on the slope and the F-test agree in simple regression.
- Read a residual plot for curvature and for a fan; measure a fan with the residual standard deviation by group.
- Compute lag correlations of residuals in time order.
- Build a Q-Q plot by hand and run the Jarque–Bera test.
- Say what each failure breaks: the slope, or only its standard error.
- Refit without unusual points and report how much the slope moves.

Next week: more than one regressor. Holidays, pollution and the day of the week go into the Ecobici model together.

Go to the exam practice →

Practice for this week

Weekly exercise set: `W09_exercises.pdf`, with solutions in `W09_solutions.pdf`.

You can already solve exercises 1–13.

Suggested before the next class, closest to the exam:

Exercise	Topic	How
1	Residuals of earnings on education (`cps`)	R with the data
5	Residual analysis for eBay auctions (`auctions`)	R with the data
8	SST, SSE, SSR and R^2 by hand	R as a calculator
9	Jarque–Bera test from skewness and kurtosis	R as a calculator
12	Diagnostics on household spending, ENIGH	R with the data

Exam practice

Problems from this week's set. ← Back to the exam spotlight · ← Back to the summary

Exam practice 1 — Earnings and education

[P2 #4c]

`cpsemployed` holds 2,809 employed workers from the US Current Population Survey:

- `cpsemployed$earnwk` — weekly earnings, in dollars
- `cpsemployed$educ` — years of education

a. Regress `cpsemployed$earnwk` on `cpsemployed$educ`. Plot the residuals against `cpsemployed$educ`.

b. Are the residuals homoskedastic or heteroskedastic? Back your answer with the residual standard deviation for workers with 11 years or less, exactly 12 years, and 18 years of education.

Exercise | Start Over | Show Hint | Run Code

```
1
```

Exam practice 2 — Tourism in Turkey

[P2 #11]

The Association of Turkish Travel Agencies reports the number of foreign tourists visiting Turkey and tourist spending, by year. The data frame `tourism` has 19 years:

- `tourism$tourists` — foreign tourists, in millions
- `tourism$spending` — tourist spending, in billions of dollars

a. Describe the relationship between the number of tourists and spending.

b. What are the explanatory and response variables?

c. Why might we want to fit a regression line to these data?

d. Do the data meet the conditions for fitting a least squares line? Use the scatterplot, the residual plot and the histogram of the residuals.

Exercise | Start Over | Show Solution | Run Code

```
1
```

Exam practice 3 — Starbucks calories and carbohydrates

[P2 #12]

Starbucks lists only the calories of its food items. We want to predict the carbohydrates of an item from its calories. The data frame `starbucks` has 29 items:

- `starbucks$calories` — calories
- `starbucks$carb` — carbohydrates, in grams

a. Describe the relationship between calories and carbohydrates.

b. What are the explanatory and response variables?

c. Why might we want to fit a regression line to these data?

d. Do these data meet the conditions for fitting a least squares line?

Exercise | Start Over | Show Hint | Run Code

```
1
```

📝 Exam practice 4 — ANOVA with a dummy: cigarettes

[P2 #24]

`cig` holds the 50 US states plus DC in 2019:

- `cig$cigsales` — annual cigarette sales, packs per person
- `cig$producer` — 1 if the state produces more than 20 million pounds of tobacco

a. Before any regression, compute the mean of `cig$cigsales` for producer and non-producer states. How many states are in each group?

b. Regress `cig$cigsales` on `cig$producer`. Check that $\hat{\alpha}$ is the non-producer mean and $\hat{\alpha} + \hat{\beta}$ the producer mean. Why must this be so?

c. What does the slope $\hat{\beta}$ measure?

d. Test $H_0 : \beta = 0$ against $H_A : \beta \neq 0$ at the 5% level. Build the t statistic from `vcov()` first, then check with `summary()`.

e. Build the two-sample pooled-variance t statistic by hand: each group's mean and variance, the pooled variance, the standard error, the p-value from `pt()`. Compare with (d).

f. Draw side-by-side boxplots of `cig$cigsales` by `cig$producer`. Do the groups look roughly normal with equal spread?

Exercise | Start Over | Show Hint | Run Code

```
1
```

📝 Exam practice 5 — eBay auctions: residual analysis

[P2 #25]

`auc` holds 684 eBay auctions of iPod minis, June–July 2006:

- `auc$finalprice` — final price, in dollars
- `auc$bidders` — number of bidders
- `auc$log_price` — natural log of `auc$finalprice`

a. Regress `auc$finalprice` on `auc$bidders` and write the fitted equation.

b. Draw and discuss: (i) residuals against fitted values; (ii) a histogram of the residuals; (iii) a Q-Q plot of the residuals, built by hand; (iv) residuals against `auc$bidders`.

c. Which assumptions — linearity, normality, constant variance — look satisfied, and which look violated?

d. Find the 5 auctions with the largest residuals in size. What might explain them?

e. Bonus: regress `auc$log_price` on `auc$bidders` and repeat the residual plot and the Q-Q plot. Does the log help?

Exercise | Start Over | Show Hint | Run Code

```
1
```

📝 Exam practice 6 — Two exams: inference and diagnostics

[P2 #26]

`ex` holds 77 university students:

- `ex$exam1` — score on the first exam, out of 100
- `ex$exam2` — score on the second exam, out of 100

a. Draw the scatterplot of `ex$exam2` against `ex$exam1`. Does it look linear?

b. Regress `ex$exam2` on `ex$exam1` and write the equation. If a student scores 10 points more on exam 1, how much higher do we expect exam 2?

c. At the 1% level, test $H_0 : \beta_1 = 0$ against $H_A : \beta_1 \neq 0$: test statistic, p-value, conclusion.

d. Report R^2. What share of the variation in exam 2 is not explained by exam 1?

e. A student scored 85 on exam 1. Give (i) a point prediction of exam 2; (ii) a 95% confidence interval for the mean exam 2 score of all students with 85; (iii) a 95% prediction interval for this student.

f. Plot the residuals against the fitted values. Are linearity and constant variance reasonable?

g. Build a Q-Q plot of the residuals by hand, and test H_0: the residuals are normal against H_A: they are not, with Jarque–Bera at 5%.

Exercise | Start Over | Show Hint | Run Code

```
1
```

Exam practice 7 — Reading residual plots

[B7]

For each description, name the assumption in trouble and say what it does to your results.

a. Residuals form a clear U-shape against the fitted values.

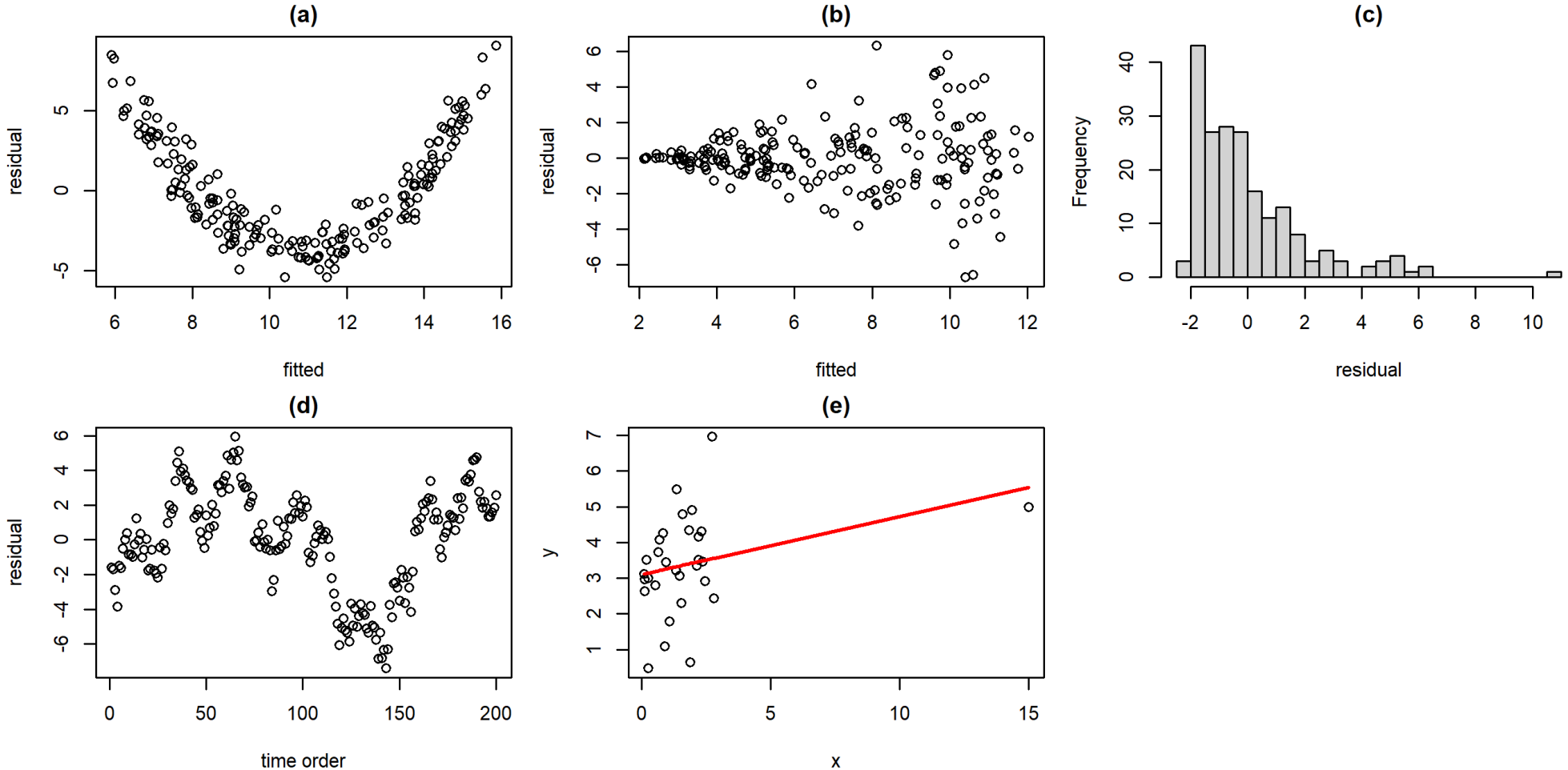
b. The vertical spread of the residuals widens steadily as the fitted values grow.

c. A histogram of the residuals has a long right tail and several extreme values.

d. Residuals plotted in time order show long runs above zero, then long runs below.

e. One observation sits far to the right on the x-axis and pulls the fitted line towards it.

One example of each:


(a)
residual
fitted
(b)
residual
fitted
(c)
Frequency
residual
(d)
residual
time order
(e)
y
x

Exam practice 8 — Sums of squares, R^2 and F

[B8]

A simple regression on $n = 25$ observations gives $SST = 4800$ and $SSE = 1200$.

a. Compute SSR.

b. Compute R^2 and say what it means in words.

c. Compute the adjusted R^2, $\bar{R}^2 = 1 - (1 - R^2)\dfrac{n-1}{n-2}$.

d. Compute $s^2 = SSE/(n-2)$ and the residual standard error s.

e. Test $H_0 : \beta_1 = 0$ against $H_A : \beta_1 \neq 0$ with the F statistic, using the 5% critical value $F_{1,23} = 4.28$.

f. A colleague says: "$R^2 = 0.75$, so the model is correct." Give a one-sentence objection.

Exercise | Start Over | Show Solution | Run Code

```
1
```

Exam practice 9 — Jarque–Bera by hand

[B9]

A regression with $n = 100$ has residuals with sample skewness $S = 0.6$ and excess kurtosis $K = 1.2$. Test normality with

$$JB = \frac{n}{6}\left(S^2 + \frac{K^2}{4}\right).$$

a. State H_0 and H_A.

b. Compute JB.

c. The statistic follows a χ^2 distribution. How many degrees of freedom, and why?

d. Compare with the 5% critical value and give the p-value. What do you conclude?

e. The sample is large. Why are non-normal residuals less of a worry here than with $n = 20$?

Exercise | Start Over | Show Solution | Run Code

```
1
```

Exam practice 10 — Non-constant variance

[B10]

A regression of household food spending on income gives residuals whose standard deviation, within each income quartile, is

$$1{,}279 \qquad 1{,}535 \qquad 1{,}931 \qquad 3{,}339$$

from the poorest to the richest quarter.

a. What is this pattern called?

b. Are the coefficient estimates still unbiased? Explain.

c. What exactly does it break, and why does that matter for the t-tests you report?

d. Name two things you could do about it.

e. Why would a manager who reads only the coefficient, never the standard error, not notice the problem?

Exercise | Start Over | Show Solution | Run Code

```
1
```

Exam practice 11 — Reading slopes after logs

[B11]

Three models are fitted to the same data. Interpret the slope in each.

a. $\widehat{\text{sales}} = 120 + 3.4\,\text{price}$ (both in pesos).

b. $\widehat{\log(\text{sales})} = 4.2 - 0.27\,\text{price}$.

c. $\widehat{\log(\text{sales})} = 1.8 + 0.51\ \log(\text{income})$.

d. Which of the three gives an elasticity? What does it mean for a 10% rise in income?

e. Give one reason, other than interpretation, why taking logs of a right-skewed variable often improves a regression.

Exercise | Start Over | Show Hint | Run Code

```
1
```

Exam practice 12 — Diagnostics on household spending (ENIGH)

[C7]

`H` holds 74,203 Mexican households (ENIGH 2018) with positive income and positive food spending, in pesos per month:

- `H$monthly_income` — household income
- `H$food_spending` — household food spending
- `H$log_income`, `H$log_food` — their natural logs

a. Regress `H$food_spending` on `H$monthly_income`. Report the intercept, the slope and R^2.

b. Interpret the slope: how much more does a household spend on food for each extra peso of income?

c. Plot the residuals against the fitted values. Describe the cloud.

d. Compute the residual standard deviation in each quartile of `H$monthly_income`.

e. Regress `H$log_food` on `H$log_income`. Report the slope and R^2, and read the slope as an elasticity.

f. Repeat (d) for the log model. Did the logs fix the problem? Which model would you report?

Exercise | Start Over | Show Solution | Run Code

```
1
```

Exam practice 13 — Diagnostics on mtcars

[C8]

`mtcars` ships with R. It holds 32 cars:

- `mtcars$mpg` — fuel economy, miles per gallon
- `mtcars$wt` — weight, in thousands of pounds
- `mtcars$log_mpg` — natural log of `mtcars$mpg`

a. Regress `mtcars$mpg` on `mtcars$wt`. Report the intercept, the slope, R^2 and the residual standard error.

b. Interpret the slope for a car that is 1,000 lb heavier.

c. The intercept is 37.3. Why does it have no sensible physical meaning?

d. Plot the residuals against the fitted values. Find the three cars with the largest residuals in size.

e. Two of those three are small, light, very efficient cars. What does that say about the straight line at the light end?

f. Regress `mtcars$log_mpg` on `mtcars$wt` and compare R^2 with (a). Which fits better, and what does the slope mean now?

Exercise | Start Over | Show Hint | Run Code

```
1
```

Appendix — optional

The algebra behind the F-test.

Appendix — why the F-test works

Whether $\beta_1 = 0$ or not, the error mean square estimates σ^2 without bias:

$$E\left(\frac{SSE}{df_E}\right) = E\left(\frac{\sum_i \hat{u}_i^2}{n-2}\right) = \sigma^2, \qquad \frac{SSE}{\sigma^2} \sim \chi^2_{n-2}$$

Only if the null is true ($\beta_1 = 0$):

$$E\left(\frac{SSR}{df_R}\right) = E\left(\frac{\sum_i (\hat{y}_i - \bar{y})^2}{1}\right) = \sigma^2, \qquad \frac{SSR}{\sigma^2} \sim \chi^2_1$$

So under the null the ratio of the two, each divided by its degrees of freedom, is

$$F = \frac{SSR/df_R}{SSE/df_E} \sim F_{1,\, n-2}$$

If the alternative is true,

$$E(SSR) = \sigma^2 + \beta_1^2 \sum_i (x_i - \bar{x})^2$$

so the numerator tends to be larger than the denominator and F is large. When the model explains y well, the explained part outweighs the unexplained part.

← Back to The F statistic